Dynamics of High-Growth Young Firms and the Role of Venture Capitalists

by

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Abstract

Motivated by the substantial growth and upfront investments of venture capital (VC) backed firms observed in administrative US Census data, this paper develops a firm dynamics model over the life cycle. In the model, startups choose the source of financing from VC, Angel investors, or banks, depending on their growth potential, and invest in innovation. The calibrated model explains the life-cycle dynamics of firms with different sources of financing and implies that venture capitalists' advice accounts for around 22% of the growth of VC-backed firms. A counterfactual economy without VC financing would lose aggregate consumption by around 0.4%.

Keyword: Venture capital, firm dynamics, innovation, upfront investment, defaultable debt, endogenous sorting

JEL Classification: D22, D25, E22, G24, G30, O32

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1 Introduction

Young firms are drivers of job creation, business dynamism, innovation, and economic growth. In particular, a small fraction of high-growth firms contribute disproportionately to the economic dynamism.[1] However, startup founders with an intention to grow their business often lack the financial resources and management skills to execute their business ideas. Venture capital (VC) is a prominent example of external financing for high-growth firms, backing more than 30% of firms with an initial public offering (IPO) since 1980, and is known for its equity investment in early-stage startups and mentorship.[2] This paper examines the role that external financing, particularly VC financing, plays in capitalizing young firms and fostering high-growth firms.

This study documents motivational facts using administrative US Census data and proprietary VC datasets: VC-backed firms achieve substantial employment and payroll growth relative to firms financed by other sources, while they require significant upfront investments. To explore the factors driving these facts, this paper develops a quantitative firm dynamics model over the life cycle, where firms are endowed with a heterogeneous growth potential, choose the source of financing, and invest in innovation. The model implies that high-potential firms self-select into VC financing to benefit from equity-based funding and managerial advice. A micro-level counterfactual simulation implies that relative to Angel financing (i.e., non-VC-equity financing), advice from venture capitalists (VCs) increases the growth of VC-backed firms by around 22%.[3] In the steady state, a counterfactual economy without VC financing would suffer aggregate output and consumption losses of around 0.2% and 0.4%, respectively.

The empirical analysis is based on the Longitudinal Business Database at the US Census Bureau. The data were merged with proprietary VC datasets via name and address matching. This dataset facilitates a comprehensive study of VC-backed firms in the US economy. At the macro level, VC-backed firms contributed to 10.6% of net employment growth and 15.8% of net payroll growth for the period from 1990 to 2019. This oc-

[1] Decker et al. (2014) demonstrates that newborn firms and high-growth firms account for around 20% and 50%, respectively, of gross job creation within the US economy. Other relevant studies include Haltiwanger et al. (2016), Akcigit and Kerr (2018), Sterk et al. (2021), and Acemoglu et al. (2022).

[2] Venture capital is defined as "independent, professionally managed, dedicated pools of capital that focus on equity or equity-linked investments in privately held, high growth companies" (Gompers and Lerner, 2001). Venture capitalists raise funding primarily through institutional investors, such as pension funds, in the form of limited partnerships and actively monitor and advise the firms in which they invest. They typically serve as board directors and recruit executives occasionally. See Metrick and Yasuda (2021) and Greenwood et al. (2022b) for the history and characteristics of VC. Examples of VC-backed firms include Amazon, Apple, Facebook, Google, Microsoft, Moderna, Netflix, Starbucks, Tesla, and Uber (Source: SDC New Issues).

[3] Angel investors are wealthy individuals who invest their own money. They may advise firms but are generally less active than VCs in monitoring and advising firms (Fenn et al., 1997). Angel investors do not include friends or relatives (Wetzel Jr, 1983).

curred even though only around 0.2% of all firms in the economy secured VC financing, showing the disproportionate importance of VC-backed firms in the economy.

At the micro level, 22% of VC-backed firms established from 1980 to 2010 became high-growth firms, while less than 1% of all firms achieved high growth. High-growth firms, so-called Gazelles, are defined as firms whose annual average employment growth rate exceeds 20% in the first five years and whose employment level exceeds 100 within 20 years.[4] In the sample of Gazelles, the growth of VC-backed firms is exceptional. The average firm size of VC-backed Gazelles at age 15 is more than five times larger than that of non-VC-backed Gazelles.

Furthermore, the data show substantial upfront investments by VC-backed firms and highlight the cost of VC financing. VC-backed firms typically raise funding that is more than 10 times their revenue at age 0, as opposed to Angel-backed firms raising 3.4 times their revenue. Angel investors are high-net-worth individuals who directly invest in early-stage companies. Research and development (R&D) intensity of VC-backed firms, measured by R&D expenditures over firm revenue, is 14% higher than that of non-VC-backed firms and declines over firm age. These two facts indicate that, compared with non-VC-backed firms, VC-backed firms raise significantly more funding relative to their revenue and invest more heavily in innovation during their initial stages. Finally, VC investors acquire 3.3% extra equity stakes compared to Angel investors in observationally comparable financing deals. This disparity can be interpreted as compensation for managerial advice from VCs. Overall, micro-level evidence suggests that the life-cycle dynamics of VC-backed firms differ significantly from those of non-VC-backed firms. The quantitative model aims to explain these patterns while providing insights into the emergence of Gazelle firms in the economy.

On the basis of descriptive evidence, this study develops the quantitative firm dynamics model with three key elements. First, unlike a firm dynamics model with an exogenous productivity process, firms invest in innovation to realize their heterogeneous growth potential. In other words, the firm's potential does not materialize without innovation investment. Second, startups choose their financing source from VC, Angel investors, and banks. Financial frictions arise from bank default costs and costs of raising equity. Because high-potential firms invest intensively in innovation and would stand a higher chance of defaulting on bank loans, these firms choose equity financing over bank financing. Third, VCs provide managerial advice to enhance the chance of successful innovation.[5] Because VC financing requires management fees paid in the form of an extra equity stake, startups seek such financing only if the benefits of the VCs'

[4]Alternatively, Gazelles can be defined by firm payroll. In the data, 36% of VC-backed firms became payroll Gazelles. Section 2 shows that Gazelles comprise more than half of the net aggregate growth in employment and payroll. The definition of Gazelles in this paper closely follows that of Sterk et al. (2021).

[5]The model assumes that the only difference between VCs and Angel investors is that VCs provide man-

advice outweigh the cost of losing an additional equity stake. The value-adding effect is calibrated to match the extra equity stake VCs obtain relative to Angel investors. To my knowledge, this paper provides the first quantitative framework for explaining firm growth over the life cycle that embeds the three interconnected factors: innovation, financing choice, and the role of VCs. As a result, the life-cycle dynamics of VC-backed, Angel-backed, and bank-backed firms are explained. The importance of the source of financing for firm growth is quantified by counterfactual exercises.

In the model, firms are endowed with heterogeneous growth potential and invest in innovation to move up the inherent productivity ladders. At age zero, firms earn zero revenue, but they must pay for labor, capital, and innovation expenditures, implying the need for external financing. As stated, firms choose VC, Angel investors, or bank financing, depending on their inherent growth potential. While the model assumes complete information on growth potential, the firm growth is subject to innovation outcomes and ex-post productivity shocks.

In bank financing, firms raise defaultable short-term debt each period and roll over the debt if necessary. The bank loan rate is endogenously determined to satisfy the zero expected profit condition of banks based on the probability of debt being repaid and potential default costs. In equity (VC or Angel) financing, firms raise funds sequentially in exchange for financiers' ownership stakes. The investors' equity stake is determined by a ratio of the money invested to the firm's overall value. While equity financing has no default risks associated with it, operating a business as an equity-backed firm incurs fixed costs every period.

VC financing differs from Angel financing in that VCs provide managerial advice and improve the probability of moving up the productivity ladder, while there is a limited supply of VCs in the economy. In exchange for their advice, VCs demand an extra equity stake. Their compensation is determined in equilibrium so that firms at the threshold are indifferent to VC and Angel financing. The additional 3.3% equity stake that VCs demand reflects their value-adding effect, as VC-backed firms may choose to raise Angel financing instead if the compensation is higher than the perceived benefit of managerial advice.

The model results in a monotonic relationship between the potential of a firm and the financing mode. Most firms borrow from banks as their default rate is relatively low, and the fixed costs of equity financing outweigh the benefits. However, bank financing is unsuitable for high-potential firms because of their large innovation expenditures and uncertain innovation outcomes. As firms borrow more from banks, the probability of

agerial advice and assumes no difference between VCs and Angel investors in their ability to screen startups and secure funding. Despite the simplifying assumptions, the quantitative model reasonably explains the dynamics of VC-backed firms in several dimensions without explicitly targeting them in calibration.

bankruptcy and associated bankruptcy costs would increase. The top 0.2% of newly established firms raise VC financing as they benefit the most from the advice of VCs due to the high return associated with successful innovation.

The calibration targets the salient features of firm dynamics, such as the average firm size over the life cycle, and identifies the innovation function, where the probability of successful innovation depends on innovation expenditures and other firm characteristics. In practice, VC-backed firms typically seek to raise multiple rounds, and reaching a subsequent round indicates successful business performance. The model targets the probability of raising subsequent VC rounds observed in the data, which are informative regarding the innovation success rate. This approach identifies the innovation function without relying on R&D expenditure or patent filing data, which take positive values for only a small fraction of all firms.[6]

The calibrated model explains the substantial growth of VC-backed firms and their upfront investments as untargeted moments. Furthermore, it reasonably matches the evolution of equity stakes owned by investors and the total money raised by VC-backed and Angel-backed firms over the life cycle without explicitly targeting them in calibration.

The calibrated model quantifies the role of VCs at the micro and macro levels. At the micro level, counterfactual growth paths of VC-backed firms are simulated under alternative financing sources. The counterfactual simulation implies that the advice of VCs enhances the expected firm size of VC-backed firms at age 10 by 22% relative to Angel financing. Conversely, the expected firm size of VC-backed firms would decline by 53% with bank financing relative to Angel financing, demonstrating the importance of equity financing for high-growth firms. At the macro level, the role of VCs in the economy is quantified by simulating an economy without VC financing. The counterfactual experiment predicts a decline in aggregate output and consumption in the steady state by approximately 0.2% and 0.4%, respectively. The findings indicate that the impact of VC financing on the economy is sizable but not as large as estimated by previous studies.[7] Finally, doubling the number of VCs in the economy, assuming that they provide the same quality of managerial advice, enhances aggregate output and consumption by 0.1% and 0.15%, respectively, as more firms receive VC advice while paying lower compensations.

[6]Only around 3% of US firms are active in R&D, according to Bessen and Wang (2024). Less than 1% of all firms generate patents each year, according to the PetentsView database. The innovation expenditures in the model aim to capture broader innovation activities.

[7]For instance, Akcigit et al. (2022) estimate that an economy without VC would have a 0.5 percentage point lower economic growth rate and suffer a welfare loss of 7.58%, measured by a compensating variation. The large welfare loss is partly due to technological spillovers from VC-backed firms to the rest of the economy. This study does not take a stance on spillover effects but closely examines the life cycle dynamics of VC-backed firms, whereas Akcigit et al. (2022) analyze two phases of firms' life cycle (i.e., startups and production firms).

Related Literature This paper contributes to four strands of the literature. First, this paper is closely related to recent studies of VC in the macroeconomy. For instance, Jovanovic and Szentes (2013), Akcigit et al. (2022), and Greenwood et al. (2022a) examine how VC-backed firms outperform other firms in the macroeconomy. Compared to their work, this study embeds VC financing within a firm dynamics model and explains the growth trajectories of VC-backed firms over the life cycle. Furthermore, this study models the endogenous choice of securing financing from VC, Angel, and bank financing and quantifies the importance of external sources of financing for firm growth.[8]

Second, the quantitative model builds on a substantial body of firm dynamics literature in the style of Hopenhayn (1992). The analysis echoes three themes in the literature: (i) firm dynamics over the life cycle of firms (e.g., Sterk et al., 2021), (ii) financial frictions due to costs of defaulting on bank loans and raising equity (e.g., Cooley and Quadrini, 2001),[9] and (iii) the productivity ladder (Cole et al., 2016). Compared to Sterk et al. (2021), this study tracks the life cycle dynamics of firms with various financing sources. Compared to Cooley and Quadrini (2001), this analysis allows for a synergy effect between VCs and entrepreneurs. In addition, unlike the work by Cole et al. (2016), the probability of moving up the ladder is endogenous and is affected by the source of financing.

Third, this work builds on empirical VC financing literature. Several studies estimate the treatment effect of VC (e.g., Kortum and Lerner, 2000; Sørensen, 2007; Bernstein et al., 2016). In comparison, this study quantifies the value-adding effect of VCs relative to Angel investors using a new calibration strategy based on the extra equity stake of VC investors compared to Angel investors.[10] Furthermore, this study measures the growth of VC-backed firms in terms of employment and payroll, which are relevant to the welfare of households.

Finally, this study contributes to the empirical literature on firm dynamics that analyzes US Census data (e.g., Haltiwanger et al., 2013; Pugsley and Şahin, 2019; Salgado et al., 2019; Ma, 2022). The study conducted by Puri and Zarutskie (2012) was the first study to document the growth of VC-backed firms using US Census data. Compared to their study, this study documents the substantial growth of VC-backed firms, even among high-growth firms, and the significant upfront investments by VCs. Moreover, this study provides the first evidence of (i) the contribution of VC-backed firms to net aggregate

[8] In Jovanovic and Szentes (2013), entrepreneurs choose between going solo or VC financing. In Akcigit et al. (2022) and Greenwood et al. (2022a), the choice of VC or non-VC is exogenous.

[9] Recent work on firm dynamics with financial frictions includes studies by Fernández-Villaverde (2010), Buera and Shin (2013), Khan et al. (2014), Midrigan and Xu (2014), Salgado (2020), Sui (2020), Corbae and D'Erasmo (2021), and Kochen (2022).

[10] Hsu (2004) finds that entrepreneurs are more likely to accept financing offers from more reputable VCs even with a discount on the company's valuation, suggesting the extra value of reputable VCs over less-reputable VCs.

growth of employment and payroll and (ii) the high R&D intensity of VC-backed firms. This finding adds to the understanding of VC-backed firms in the sample of all employer businesses in the United States.

2 Descriptive Evidence

This section presents three sets of descriptive evidence: VC-backed firms exhibit high growth in employment and payroll, VC-backed firms make large upfront investments in their early life, and VC investors acquire more equity stakes than Angel investors in observationally comparable financial deals. Comparisons between VC-backed and non-VC-equity (Angel) backed firms are particularly novel, as little information on Angel-backed firms has been documented in the literature. The facts presented here motivate the quantitative firm dynamics model with VC financing and innovation.

2.1 Data

The primary source of firm-level microdata is the Longitudinal Business Database (LBD), which is collected and organized by the US Census Bureau.[11] The LBD covers virtually the entire population of employer firms and contains information on employment, payroll, revenue, firm age, industry (NAICS code), and firm death. The sample spans from 1980 to 2019 in this study. Firm size is primarily measured in employment and payroll. Revenue data have been added to the LBD since 1997. The LBD is linked to the Census Bureau's Business Register, which contains the name and address of all establishments.

The Survey of Business Owners (SBO) in 2007 supplements the information on the source of financing. The survey asks more than one million businesses in the United States about their sources of capital, such as early-stage equity investments and business loans.[12] The data on R&D expenditures are drawn from the Survey of Industry Research and Development (SIRD) for the period 1980–2007 and from the Business R&D and Innovation Survey (BRDIS) for the period 2008–2019. These datasets are a part of the restricted-use database of the Census Bureau and are linked to the LBD with census identifiers.

This study implemented name and address matching between the Business Register and the four external VC datasets: Crunchbase, Pitchbook, Preqin, and Refinitiv's VentureXpert. The VC datasets provide information on the size and timing of most VC deals in the sample and the equity share of investors for a subset of VC deals. Crunchbase

[11] Appendix A.1 describes each dataset more in detail.

[12] In this study, non-VC-equity-backed firms are defined as firms that raised equity financing but did not raise VC financing. Bank-backed firms are defined as firms that received bank loans but did not raise equity financing.

and Pitchbook contain information on some Angel deals, which enables comparisons between VC deals and Angel deals.[13] In addition, two datasets, the SDC New Issues and Ritter's IPO dataset, are merged to supplement the VC-backing status of firms that eventually went public.

2.2 Growth Accounting

This study conducts an accounting exercise to measure the contribution of VC-backed firms and high-growth firms to net aggregate growth, exploiting the sample of essentially all firms in the US economy.[14]

High-growth firms, which are often called "Gazelles" in the literature, are defined based on the firm's employment or payroll. Employment Gazelles are firms whose average annual employment growth rate is above 20% in the first five years and whose employment exceeds 100 by age 20. Payroll Gazelles are defined likewise.[15] VC-backed firms are those perceived to raise VC financing in the VC datasets and are linked to firms in the LBD. For the period from 1980 to 2019, around 0.2% of all firms are VC-backed, and around 0.8% of all firms are Gazelles, as shown in Table 1.[16]

The contributions of VC-backed firms and Gazelles to net aggregate growth are computed in several steps. The mathematical definitions are provided in Appendix A.2. First, the growth rate of all firms is computed as a weighted sum of the growth rates of establishments belonging to the firms in the next period, following Davis et al. (1996) (hereafter, DHS). The advantage of this approach is that it measures the organic expansion of

[13] Since Angel investments are undertaken by individual investors, academic researchers generally face challenges in analyzing comprehensive data on Angel financing. Crunchbase and Pitchbook also cover only a subset of Angel deals. To the extent that financing deals by reputable Angel investors are more likely to be included in the data than other Angel deals, the difference between VC deals and Angel deals in this study is underestimated.

[14] The importance of VC-backed firms in the aggregate economy has been documented in the literature (cf. Greenwood et al., 2022a). However, the literature typically examines publicly traded firms, which account for a relatively small share of the total employment. According to Dinlersoz et al. (2018), the domestic employment of actively-traded firms accounts for around 26% of the total private employment in the United States.

[15] Payroll Gazelles are firms with an average annual payroll growth rate above 20% in the first five years and with their payroll exceeding 100 times the average real wage per worker in the economy by age 20. Sterk et al. (2021) define Gazelles as "startups with an ex ante projected growth rate of at least 20 percent annually, over the first five years, and an expected employment level of at least 10 workers at some point during their lifetimes." In their definition, 5.4% of all startups are Gazelles. Here, the criteria is set to 100 employees instead of 10 employees to narrow the focus on high-growth firms. Note that the definition of Gazelles in Birch (1987) is based on revenue, but revenue is not observable for all firms in the LBD.

[16] The definition of Gazelles is based on the firm growth rate from age 0 to 5 and firm size up to 20 (unless the firm reaches employment of 100 or above before age 20). Therefore, some of the firms established before 1980 or after 2000 could be misclassified as non-Gazelle firms. The growth rate in the first five years for the firms established before 1980 is computed based on the growth rate from 1980 to 1985. Table 2 reports the fraction of Gazelles in the sample of firms established from 1980 to 2010, which is close to the fraction reported in Table 1.

Table 1: Contribution of VC-backed Firms and Gazelles to Net Aggregate Growth

	Fraction (%)	Employment growth (%)	Payroll growth (%)
VC-backed	0.2	10.6	15.8
"Gazelles"	0.8	65.2	51.3

*N*ote: Employment Gazelles are firms with an average annual growth rate above 20% in the first five years and employment of 100 or above by age 20. Payroll Gazelles are defined likewise. The fraction of employment Gazelles and payroll Gazelles is 0.793% and 0.788%, respectively, both of which are approximately 0.8%. The fraction of firms is computed for the period from 1980 to 2019. The contribution to aggregate growth is for 1990–2019. Data: LBD (1980–2019) and VC datasets. FSRDC Project Number 2633 (CBDRB-FY23-P2633-R10843).

firms, which is not affected by ownership changes. Based on the DHS firm growth rates, the growth rate of a group of firms (i.e., VC-backed firms or Gazelles) is computed as a weighted sum of the firm growth rates within the group. Then, the contribution of the group of firms to aggregate growth is computed in each year. Finally, the contribution of the groups of firms from 1990 to 2019 is computed based on the geometric mean of the growth rates.

The results of the growth accounting are summarized in Table 1. Despite their small fraction in the economy, VC-backed firms account for 10.6% of net employment growth and 15.8% of net payroll growth from 1990 to 2019.[17] Gazelles account for 65.2% of net employment growth and 51.3% of net payroll growth, demonstrating their disproportionate contributions. In terms of payroll growth, VC-backed and Gazelle firms contribute approximately 80 and 65 times more to aggregate growth, respectively, than typical firms in the economy. Note that VC-backed firms account for a higher share of payroll growth than that of employment growth, implying that VC-backed firms create more high-paying jobs.

2.3 High Growth of VC-backed Firms

The firm's growth at the micro level is analyzed in the sample of firms established from 1980 to 2010. Approximately 18.2 million firms were created during this period (Table 2). 27,500 firms are identified as VC-backed firms. Thus, around 0.15% of newborn firms are backed by VCs in the sample.[18] Employment Gazelles and payroll Gazelles account for 0.86% and 0.88% of all newborn firms, respectively. In contrast, 21.8% and

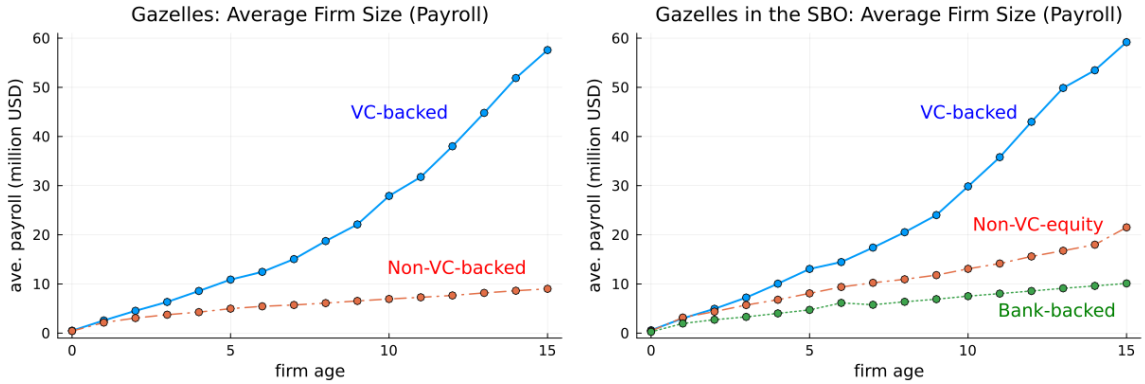
[17]Compared to the literature, Puri and Zarutskie (2012) show that the employment share of VC-backed firms is around 4–7% from 1991 to 2005. From 2017 to 2021, the employment share of VC-backed firms is around 13%, according to Ando et al. (2024).

[18]Here, the fraction of VC-backed firms is smaller than the fraction reported in Table 1, because the sample in Table 1 includes firms established after 2010. In this subsection, firms born from 1980 to 2010 are analyzed so that the growth rate from age 0 to 5 and firm size for at least 10 years are observed for all firms. Compared to the literature, Puri and Zarutskie (2012) report 17,763 VC-backed firms and 16,286,524 non-VC-backed firms born from 1981 to 2005. This implies that about 0.1% of newborn firms

Table 2: The number of newly established firms from 1980 to 2010

	Any firm	Employment Gazelles	Payroll Gazelles
All firms	18,150,000	156,000 (0.86%)	160,000 (0.88%)
VC-backed firms	27,500	6,000 (21.8%)	9,900 (36%)

*N*ote: The sample consists of firms established from 1980 to 2010. Employment Gazelles are firms with an average annual growth rate above 20% in the first five years and employment of 100 or above by age 20. Payroll Gazelles are defined likewise. The percentages in parentheses show the fraction of Gazelles. The number of firms is properly rounded following the disclosure avoidance rules of the US Census Bureau. Data: LBD (1980–2019) and VC datasets. FSRDC Project Number 2633 (CBDRB-FY23-P2633-R10186).



*N*ote: The left panel compares VC-backed firms and non-VC-backed firms in the sample of Gazelle firms established from 1980 to 2010. The right panel compares VC, non-VC-equity, and bank-backed firms, using a subset of firms in the left panel that reported the source of capital in the Survey of Business Owners (2007). Payroll is denominated in 2009 US dollars based on GDP deflators. Data: LBD (1980-2019), SBO(2007), and VC datasets. FSRDC Project Number 2633 (CBDRB-FY23-P2633-R10186).

Figure 1: Growth Trajectory of VC-backed and non-VC-backed Gazelles

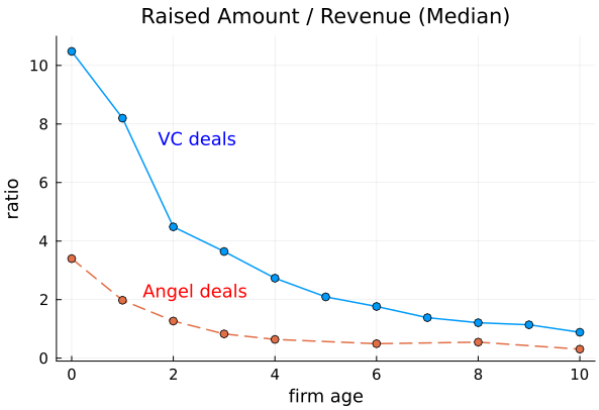
36% of VC-backed firms become employment Gazelles and payroll Gazelles, respectively. The odds are around 25 to 40 times higher than typical firms in the economy.

Next, in the sample of Gazelles, the growth trajectories of VC-backed and non-VC-backed Gazelles are compared. The left panel in Figure 1 shows the average firm size of VC-backed Gazelles and non-VC-backed Gazelles from age 0 to 15, measured in terms of payroll. VC-backed Gazelles sustain higher growth than non-VC-backed Gazelles. The average firm size of VC-backed Gazelles at age 15 is around $60 million, more than five times that of non-VC-backed Gazelles. Here, the raw sample mean of firm size is documented. The Deaton-Paxson regression (Figure 12) in Appendix A.3 shows that the difference in growth trajectories is not mainly driven by year or industry effects.[19]

The right panel compares the growth trajectories of VC-financed, non-VC-equity-financed, and bank-financed Gazelles in the sample of firms that appear both in the LBD

are VC-financed. Here, the fraction is slightly different due to three reasons: (a) more firms received VC financing after 2005, (b) the four VC datasets are examined in this study instead of two datasets in their study, and (c) I partially address the issue of spurious new firms, as described in Appendix A.1.

[19]Furthermore, regression Table 10 in Appendix A.3 shows that the DHS firm growth rate, which is robust to survival bias, is higher for VC-backed Gazelles than non-VC-backed Gazelles, conditional on the



*N*ote: The raised amount (i.e., deal size) is drawn from the VC datasets. Angel-backed firms are defined as firms that raised Angel financing, according to Pitchbook and Crunchbase, but have not raised VC financing. Revenue is from the LBD (1997-2019). FSRDC Project Number 2633 (CBDRB-FY23-P2633-R10843).

Figure 2: Upfront Investment of VC-backed and Angel-backed Firms

and in the SBO. The information on the source of capital is drawn from the SBO (2007). The average growth of VC-backed Gazelles is higher than that of non-VC-equity-backed Gazelles, followed by that of bank-backed Gazelles.[20] This monotonic relationship between the source of financing and the average growth of Gazelles will be explained with the quantitative model.

The descriptive evidence about the high growth of VC-backed firms is documented here without showing a causal relationship. However, the quantitative model with several channels at play explains the high growth of VC-backed firms. The model will also be used to predict the growth path of firms with an alternative source of financing. Note that Ando et al. (2024) documents a positive treatment effect of VC financing on firm size using a difference-in-differences approach with matched control samples. Bernstein et al. (2016) shows that on-site involvement by VCs has a positive impact on firm outcomes, exploiting the introduction of new airline routes as a source of variation.

2.4 Upfront Investment

This subsection demonstrates upfront investments of VC-backed firms. The first measure of upfront investments is the ratio of the size of funding raised in VC or Angel financing to the firm's revenue. Figure 2 shows the median ratio in the sample of VC-backed

initial year and NAICS code. Figures 11 and 12 in Appendix A.3 show the growth of VC-backed firms in employment and revenue. A comparison of exit rates is shown in Table 11. VC-backed firms have a lower exit rate than non-VC-backed firms in the entire sample and a slightly higher exit rate than non-VC-backed firms in the sample of Gazelles.

[20]Table 9 in Appendix A.1 shows the monotonic relationship between the source of financing and the odds of being Gazelles.

Table 3: R&D/revenue of VC-backed and non-VC-backed firms

	Dependent: R&D/revenue (%)	
	(1)	(2)
	all firms	payroll Gazelles
VC-backed dummy	14.5 ***	14.2***
	(0.743)	(0.882)
firm age	–0.0629***	–0.0788***
	(0.0137)	(0.0204)
VC dummy × firm age	–0.347***	–0.306***
	(0.0391)	(0.0459)
log(employment)	–1.46***	–1.24***
	(0.0684)	(0.121)
Year fixed effects	Yes	Yes
NAICS (4 digits)	Yes	Yes
N	241,000	80,000
R^2	0.329	0.436

Standard errors in parentheses

* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

*N*ote: The sample in column (1) consists of firms with positive R&D. The sample in column (2) consists of payroll Gazelles with positive R&D. The data on R&D expenditures are taken from the SIRD (1980-2007) and BRDIS (2008-2019) datasets. Revenue is drawn from the LBD. If it is unavailable, sales in the SIRD and BRDIS are used. R&D/firm revenue is winsorized at 5%. Observations are weighted by the survey weights in the SIRD and BRDIS. The sample period is from 1980 to 2019. FSRDC Project Number 2633 (CBDRB-FY23-P2633-R10843).

and Angel-backed firms from age 0 to 10.[21] The plot demonstrates that VC-backed firms typically raise more than 10 times their revenue at age 0. The funding size of VC deals consistently exceeds the revenue of the firm until age 9. Angel-backed firms also raise funding larger than their revenue at age 0, but the ratio is smaller than 1 after age 3. Therefore, VC-backed firms make significant upfront investments in their early life compared to Angel-backed firms. Note that such large upfront investments cannot be explained with a standard firm dynamics model that does not consider innovation activities.

Furthermore, this study documents the upfront investments by VC-backed firms using the data on R&D expenditures, which is a commonly used measure of innovation inputs. The R&D data are drawn from the SIRD/BRDIS datasets at the US Census. The two datasets cover a large sample of firms (approximately 25,000–40,000 firms each

[21]To avoid disclosing confidential information, the pseudo median, which is the mean of a subset of observations along the median, is reported following the guidelines of the Census Bureau. In the sample of Angel-backed firms, observations at ages 4-5, 6-7, 8-9, and 10-11 are combined to secure a reasonable sample size.

Table 4: Comparison between VC deals and Angel deals

	Dependent variable:		
	(1)	(2)	(3)
	log(deal size)	equity share (%)	deal size/revenue
VC deal dummy	1.448***	3.285***	7.232***
log(employment)	0.502***	–3.873***	
log(deal size)		5.731***	
firm age			–0.994***
Year fixed effects	Yes	Yes	Yes
Age fixed effects	Yes	Yes	No
Deal Number	Yes	Yes	Yes
Initial Year	Yes	Yes	Yes
NAICS (4 digits)	Yes	Yes	Yes
N	37500	22500	37500

* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

*N*ote: Deal number is the number of rounds in VC or Angel deals. The initial year is the year when the oldest establishment of the firm reports positive payrolls for the first time. The ratio (deal size/revenue) is winsorized at 5%. Data: LBD (1997-2019) and the VC datasets. FSRDC Project Number 2633 (CBDRB-FY23-P2633-R10843).

year) and contain young firms that are not publicly traded. The R&D intensity is defined as R&D expenditures divided by firm revenue. Table 3 shows the regression results comparing the R&D intensity of VC-backed and non-VC-backed firms, based on the following regression equation:

$$y_{it} = \beta_1 \, \mathbf{1}\{\text{VC backed}_i\} + \beta_2 \, \text{age}_{it} + \beta_3 \, \mathbf{1}\{\text{VC backed}_i\} \times \text{age}_{it} + \text{controls}_{it} + \epsilon_{it},$$

where y_{it} is an R&D/firm revenue ratio of firm i in year t, and ϵ_{it} is an error term. Here, firms with positive R&D expenditures are analyzed. The table shows that VC-backed firms typically have higher R&D intensity by around 14 percentage points relative to non-VC-backed firms and that the intensity declines over firm age. A similar pattern is observed in the sample of Gazelles in column (2). Those results show that VC-backed firms intensively invest in innovation, particularly in their early life. To my knowledge, the evidence of upfront investments of VC-backed firms, measured by the money raised over firm revenue or R&D expenditures over firm revenue, is new to this study.

Table 5: Firm age at the First VC deal

	Fraction (%)	Cumulative Fraction (%)
Age ≤ 0	46.77	46.77
$1\leq$ Age ≤ 2	27.45	74.22
$3\leq$ Age ≤ 5	14.05	88.27
Age > 5	11.73	100

*N*ote: Firm age is zero in the year when the oldest establishment of the firm reports positive payrolls for the first time. Data: LBD (1980-2019) and the VC datasets. FSRDC Project Number 2633 (CBDRB-FY23-P2633-R10843).

2.5 Equity Share in VC Financing

This subsection documents the difference in equity stakes obtained by VC investors and Angel investors. Table 4 shows the results of regression analysis that compares characteristics of VC deals and Angel deals conditional on firm characteristics observed in the LBD, such as log(employment), firm age, and industry:

$$y_{it} = \beta_1 \ \mathbf{1}\{\text{VC backed}_i\} + \text{controls}_{it} + \epsilon_{it},$$

where y_{it} is log(deal size), equity stakes acquired by the investors, and deal size/firm revenue in Columns (1)–(3), respectively. Column (1) shows that the size of VC deals is typically larger than that of Angel deals. Column (2) shows that VC investors acquire around 3.3% extra equity stakes conditional on observable firm characteristics. Column (3) confirms the evidence on upfront investments documented in Figure 2: VC-backed firms raise larger financial capital relative to their revenue than Angel-backed firms, and the ratio declines over firm age. All regressions control for year, deal number, initial year, and industry (4-digit NAICS code). The evidence of extra equity stakes in column (2) can be interpreted based on the revealed preference argument that VCs' advice is valuable from the startup's perspective.[22] Figure 13 in Appendix A.3 shows the robustness of the extra equity stakes in VC deals with descriptive statistics.

Firm age at the first VC deal Table 5 reports the firm age at the first VC financing. Firm age is defined in the LBD, where firm age is zero in the year when the oldest establishment of the firm reports positive payrolls for the first time. Almost half of the

[22]An alternative interpretation is that the difference in equity stakes reflects different risks involved. However, while VC funds invest in a portfolio of companies and diversify idiosyncratic risk, Angel investors typically have a less diversified portfolio. One could also argue that VCs acquire higher equity stakes due to larger bargaining power, but this advantage may stem from the value they bring to the companies. Finally, one may be concerned about firms' qualities unobservable to researchers. If VC-backed firms have higher unobserved qualities than Angel-backed firms, the extra equity stakes in the regression are underestimated, as higher-quality firms would give up smaller equity stakes. If 3.3% extra equity stakes are underestimated, the value-adding effect of VCs predicted by the counterfactual exercise is also underestimated.

VC-backed firms obtain the first VC deal by age zero, and around three-fourths of VC-backed firms raise VC funding by age two. This evidence shows that VC-backed firms raise VC financing early in their life and motivates a model in this study in which all firms choose the source of financing at birth.

3 The Model

This study develops a firm dynamics model over the life cycle with ex-ante heterogeneity, innovation, and external financing and explains the life-cycle dynamics of firms with different financing sources. Newly established firms are endowed with heterogeneous growth potential and choose the source of external financing at birth.[23] External financing is crucial to pay for labor, capital, and innovation expenditures, where successful innovation enhances the firm's productivity. Once the firm achieves its full growth potential, its productivity depends only on exogenous productivity shocks, as in Hopenhayn (1992), and external financing plays a minimal role thereafter. The model assumes that information is complete and abstracts from informational issues, such as incomplete information and asymmetric information.[24] The economy is studied in general equilibrium.

3.1 Production

Each firm has inherent growth potential, which can be thought of as the quality of business ideas. An ambitious idea has the potential to yield high growth, but it does not materialize without innovation investment. The potential is represented by a sequence of productivities $\{\hat{\theta}_1, \cdots, \hat{\theta}_N\}$, which is called a productivity ladder. The firm achieves its full potential when it reaches the last step of the productivity ladder. The probability of moving up the ladder (i.e., successful innovation) is endogenously determined by

[23] In the model, firms make the choice at birth and cannot raise money from other sources. Practically, firms may raise money from several sources at the same time or over time. Nonetheless, Robb and Robinson (2014) show that VC-backed startups raise the majority of their funding from VC. Anecdotally, Google raised around 1 million dollars from Angel investors and 25 million dollars from VC in its initial years. The model captures the difference between VC-backed firms that primarily rely on VC financing and Angel-backed firms that do not raise VC financing at all. According to Sohl (2012), the Angel market invests in more than 10 times more ventures than the VC market, implying that most Angel-backed firms do not raise VC financing. Furthermore, Hellmann et al. (2021) find that VC financing and Angel financing are dynamic substitutes and that transitions between financing streams are not as common as traditionally assumed in the literature.

[24] This approach has limitations since venture capitalists may be better at screening startups, but quantifying such channels would be challenging. In the literature, Sørensen (2007) estimates a two-sided matching model of venture capital and distinguishes between VCs' influence and sorting under the assumption of complete information. Note that Georges Doriot, a founder of American Research and Development Corporation, which is regarded as the first successful venture capital firm, realized that project selection was the easiest part. He wrote that "the hardest task is to help a company through its growth pains" (Ante (2008), p.121).

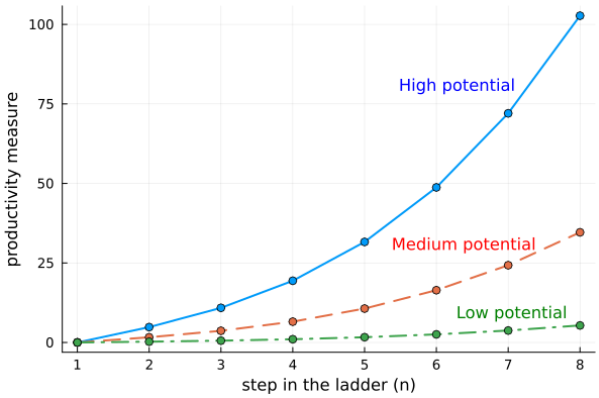


Figure 3: Examples of Productivity Ladder

innovation expenditures as well as VCs' advice and other firm characteristics.

Figure 3 depicts examples of the productivity ladder. Different productivity ladders capture heterogeneity in business ideas. In practice, some founders start their businesses with an ambitious idea and aim to expand their business. Most founders pursue a less ambitious business idea and have little intention of expanding their business.

Firms are risk-neutral and maximize the discounted sum of future expected profits. Firms produce goods with a decreasing return-to-scale technology by hiring workers and renting capital. The production function is:

$$\underbrace{y_t}_{\text{revenue}} = \underbrace{\left(\hat{\theta}_{n_t} + \epsilon_t\right)^{1-\eta}}_{\text{productivity}} \underbrace{\left(k_t^{\alpha} \ell_t^{1-\alpha}\right)^{\eta}}_{\text{production inputs}}, \tag{1}$$

where k_t and ℓ_t denote capital and labor, respectively. Capital and labor are static inputs and purchased at prices R and w, respectively. The costs are paid one period in advance before production yields revenue. A span-of-control parameter is given by $\eta < 1$.

A firm's productivity is a sum of two elements, $\hat{\theta}_{n_t}$ and ϵ_t. The first element $\hat{\theta}_{n_t}$ is determined by the firm's potential and innovation outcome. All firms start with the first step of their productivity ladder, $n = 1$, with $\hat{\theta}_1 = 0$. If their innovation is successful with probability ς, they move up to the next step. Otherwise, the firms stay at the same step of the ladder in the next period. This is formalized in the following equation:

$$n_{t+1} = \begin{cases} n_t + 1 & \text{with prob. } \varsigma_{t+1} \\ n_t & \text{with prob. } 1 - \varsigma_{t+1} \end{cases}, \tag{2}$$

where $n_t \in \{1, ..., N\}$ denotes a position in the ladder at time t. The probability of moving up the ladder ς_t is endogenously determined by innovation expenditures. The

second element ϵ_t follows an exogenous $\mathrm{AR}(1)$ process:

$$\log(\epsilon_t) = \rho \log(\epsilon_{t-1}) + \sigma_\nu \nu_t \text{ with } \nu_t \sim \text{Normal}(0,1), \tag{3}$$

where ρ is the persistence of the exogenous productivity shocks, and σ_ν is the volatility of the shocks.

The probability of moving up the ladder $\varsigma_{t+1} \in (0,1)$ is determined in the following logistic function:

$$\underbrace{\varsigma_{t+1}}_{\text{success rate}} = \frac{\exp(x_t)}{1+\exp(x_t)}, \tag{4}$$

$$\begin{aligned} \underbrace{x_t}_{\text{factor of success}} &= \gamma_0 + \underbrace{\gamma_1}_{(+)} \underbrace{\log(I_t)}_{\text{innovation expnd}} + \underbrace{\gamma_2}_{(-)} \underbrace{\log(1+\hat{\theta}_N)}_{\text{firm's potential}} \\ &\quad + \underbrace{\gamma_3}_{(-)} \mathbf{1}\{\text{fail}_t\} + \underbrace{\gamma_4}_{(+)} \mathbf{1}\{\text{VC}\} + \gamma_5 \underbrace{\log(n_t)}_{\text{step in the ladder}}, \end{aligned} \tag{5}$$

where the success rate ς_{t+1} increases with the factor of success x_t. The factor of success (x_t) is assumed to depend on six terms in equation (5) to match empirical moments discussed in Section 4.1.[25] The second term means that the success probability is increasing in innovation expenditure I_t. The third to sixth terms in equation (5) are other firm characteristics relevant to the success rate. A negative coefficient γ_2 means that more ambitious ideas with higher potential $\hat{\theta}_N$ are harder to materialize. A negative γ_3 implies that if a firm failed in innovation last period, the probability of success also declines in the coming period. In practice, startups that are not growing will have a hard time keeping their momentum.[26] A coefficient γ_4 measures the value-adding effect of VCs. If VCs provide managerial advice, firms have a higher chance of successful innovation. Finally, a coefficient γ_5 captures the relation between the success probability of innovation and the position on the ladder. All six coefficients are estimated in calibration.

Firms obtain revenue from production in the next period, as in Cooley and Quadrini

[25] If the factor of success (x_t) depends only on the second term (i.e., $\gamma_0 = \gamma_2 = \gamma_3 = \gamma_4 = \gamma_5 = 0$), innovation expenditures (I_t) can be expressed as a function of innovation success rate:

$$I_t = \left(\frac{\varsigma_{t+1}}{1-\varsigma_{t+1}} \right)^{1/\gamma_1}. \tag{6}$$

This functional form is similar to Greenwood et al. (2022a), where the innovation expenditures can be expressed as $I_t = \text{constant} \times \frac{\varsigma_{t+1}^2}{1-\varsigma_{t+1}}$. Both functional forms have the advantage of having an interior solution for innovation expenditures.

[26] For instance, Michael Seibel, managing director of Y Combinator, says about startups: "The second that you launch, every week that you're not growing, you're dying." Source: "7 Rules Every Tech Startup Must Follow," *Fortune*, November 3,2016. Cole et al. (2016) introduce the productivity ladder and assume that if a firm stalls at the current step of the ladder, it remains at the current level forever after, implying $\gamma_3 = -\infty$ in this setting. I relax this assumption and set γ_3 in calibration.

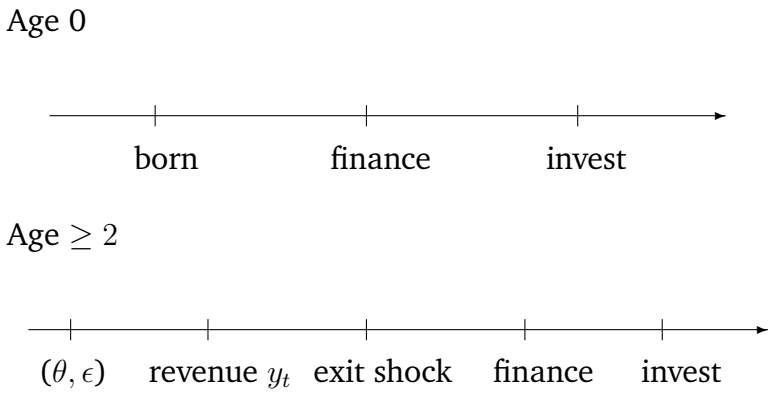


Figure 4: Timeline

(2001), and need to raise external funding if their cash flow is short of the expenditure. In practice, a critical job of startup founders is to avoid running out of cash, which is a serious concern, especially for firms that choose a path of high growth and a rapid scale-up of the business.[27] The setup for the timing of expenditure and revenue in the model captures this practical aspect.

Figure 4 shows the timeline. Firms are endowed with growth potential $\hat{\theta} := \{\hat{\theta}_n\}_{n=1,\cdots,N}$, initial exogenous productivity ϵ_0, and initial assets o_0. Given their initial state variables, firms decide the source of funding: VC financing, Angel financing, or bank financing. Then, they raise funding and pay for labor, capital, and innovation expenditures. In the first period, at age 0, firms do not collect revenue from their production.

At the beginning of period t, productivity $(\hat{\theta}_{n_t}, \epsilon_t)$ is realized. Given labor and capital inputs (ℓ_t, k_t) chosen in the previous period, the firm earns revenue according to the production function (1). After the realization of revenue, firms may be hit by an exogenous exit shock with probability φ. In this case, firms cannot continue the business and exit the market. Otherwise, firms may raise additional external funding and pay for labor, capital, and innovation expenditures. If the business is not profitable enough to operate, the firm may endogenously exit the market. This may happen as firms need to pay for operational fixed costs c_f every period. One period is set to be two years in the model, as the duration between VC deals is typically around 1.5 to 2 years. The details of the firm's decisions are discussed in the following subsections.

3.2 Equity (VC or Angel) Financing

Firms raise funding from VC investors, Angel investors, or banks. The choice of the source of financing depends on the firm's initial characteristics.

[27] See, for instance, Feld and Mendelson (2019) for issues that startup founders face in starting their business and managing financing.

In equity financing, firms raise funding from investors in exchange for an equity stake in the company. Since investors own a part of the firm, they will receive dividends once the business profits materialize in the future. Firms pay a fixed cost $c_{e,0}$ every period to operate as equity-backed firms.[28] Additionally, raising one unit of funding costs $c_{e,1}$ as flotation costs. Those two costs are the distortions of equity financing.

Venture capitalists provide managerial advice. The advice is beneficial for enhancing the probability of moving up the productivity ladder. In exchange, firms pay management fees $\bar{\pi}_{\text{vc}}$ every period to the venture capitalists.[29] The supply of VCs in the market is fixed, reflecting a shortage of human capital as in Jovanovic and Szentes (2013), and the payments to the VCs are determined in equilibrium. The compensation $\bar{\pi}_{\text{vc}}$ is reflected in the VCs' extra equity stakes. In contrast, Angel investors operate in a competitive market and make zero expected profits. Except for the advice and management fees to VCs, VC financing and Angel financing have the same structure.

Equity investors acquire the equity stake of the firm (s_t) based on the value of the financial investment relative to the firm's overall value. Firms may raise funding in several stages, and in the second or later round of equity financing, the investor's share in earlier rounds will be diluted. Taking this into account, the total equity share of investors s_{t+1} evolves according to the following formula:

$$\underbrace{s_{t+1}}_{\text{total stakes}} = \underbrace{s_t}_{\text{previous stakes}} + (1 - s_t) \underbrace{\frac{(1 + c_{e,1})(-d_t)}{V_t(n_t, \epsilon_t, o_t, \text{fail}_t; \hat{\theta}) + (1 + c_{e,1})(-d_t)}}_{\text{newly acquired stakes}}, \tag{7}$$

where $(-d_t)$ denotes the amount of funding raised in this period, and $V_t(n_t, \epsilon_t, o_t, \text{fail}_t; \hat{\theta})$ denotes the firm's value before raising money.[30]

A value function of equity-backed firms, $V_t(n_t, \epsilon_t, o_t, \text{fail}_t; \hat{\theta})$, is now formulated. The four state variables are the position of the ladder (n_t), exogenous productivity shock (ϵ_t), cash flow at the beginning of the period (o_t), which is equal to the firm's revenue in equity financing except for the initial period, and the most recent innovation outcome ($\text{fail}_t \in \{0, 1\}$). Given the state variables, firms decide whether to continue their business and, if so, choose labor and capital inputs and innovation expenditures. The revenue minus the sum of all expenditures is distributed as dividends.

[28] In practice, equity-backed firms typically register as incorporated businesses, establish a board of directors, hire lawyers and accountants, and disclose financial information to their investors. The cost of raising equity includes these costs.

[29] In reality, firms and VCs form a sophisticated contract in which payment to VCs is contingent on the firm's performance. The setup here simply captures that VCs are compensated for their managerial advice.

[30] The derivations are shown in Appendix B.1. Intuitively, the investor's share s_t is equal to the value of money invested divided by the firm's post-money valuation. The two are not exactly the same in the model because of the flotation costs and compensation for VCs. In a second or later round of equity investment, the equity share of the past equity investors will decrease to offset the increase in the firm's value.

Firms exit the market if the continuation value is smaller than their cash flow. Hence, the firm's value is the maximum of the continuation value and its cash flow:

$$V_t^j(n_t, \epsilon_t, o_t, \text{fail}_t; \hat{\theta}) = \max\{\underbrace{V_t^{j,C}(n_t, \epsilon_t, o_t, \text{fail}_t; \hat{\theta})}_{\text{continue}}, \underbrace{o_t}_{\text{cash flow}}\} \text{ for } j \in \{\text{VC}, \text{Angel}\}. \quad (8)$$

If the firm continues the business operation, the continuation value is as follows:

$$V_t^{j,C}(n_t, \epsilon_t, o_t, \text{fail}_t; \hat{\theta}) = \max_{\ell_{t+1}, k_{t+1}, I_t, d_t} \underbrace{d_t}_{\text{dividend(+) or deal size(–)}} - \underbrace{\mathbf{1}\{(-d_t) > 0\}(-d_t c_{e,1})}_{\text{cost of raising equity}}$$

$$+ \frac{1}{1+r}\left\{(1-\varphi)\underbrace{\mathbb{E}\left[V_{t+1}^j(n_{t+1}, \epsilon_{t+1}, o_{t+1}, \text{fail}_{t+1}; \hat{\theta})|\varsigma_{t+1}\right]}_{\text{no exit shock}} + \varphi \underbrace{\mathbb{E}\left[o_{t+1}|\varsigma_{t+1}\right]}_{\text{exogenous exit}}\right\} \quad (9)$$

$$\text{s.t. } d_t + \underbrace{Rk_{t+1} + w\ell_{t+1} + I_t + c_f + c_{e,0} + \mathbf{1}\{\text{j=VC}\}\bar{\pi}_{\text{vc}}}_{\text{expenditures}} = \underbrace{(\hat{\theta}_{n_t} + \epsilon_t)^{1-\eta}(k_t^\alpha \ell_t^{1-\alpha})^\eta}_{=:o_t \text{(cash flow)}} \quad (10)$$

where d_t is the value of the dividend if it is positive, and $(-d_t)$ is the amount of money raised if d_t is negative.[31] If the firm is hit by an exogenous exit shock in the next period with probability φ, the firm must exit the market with its cash flow o_{t+1}. Otherwise, the firm value is given by equation (8). In equation (10), R is the rental rate of capital, w is the wage of workers, I_t is the innovation expenditure, c_f is the fixed operational costs, and $c_{e,0}$ is the additional fixed cost for equity-backed firms. In VC financing, firms pay management fees $\bar{\pi}_{\text{vc}}$ to VCs every period. Innovation expenditures I_t determine the probability of stepping up the ladder ς_{t+1} in the next period, according to the innovation functions (4) and (5). If the sum of all expenditures is larger than revenue, firms rely on external financing.

Management fees to venture capitalists $\bar{\pi}_{\text{vc}}$ is determined in equilibrium. Since the supply of VCs is limited, only a fixed fraction of all startups can be funded by VCs. Hence, startups need to compensate for VCs' advice. While newborn firms are heterogeneous in their potential, VCs are homogeneous. Thus, all VCs obtain the same compensation. In equilibrium, all VC-backed firms compensate $\bar{\pi}_{\text{vc}}$ for VCs' advice so that the threshold firm is indifferent between VC financing and Angel financing.[32]

[31] In the model, firms never distribute dividends and raise money from investors at the same time, as external financing is more costly than internal financing. In equity financing, firms do not make a savings decision in the model. This is mainly for computational tractability, although the savings decision is less relevant for young firms, especially for firms scaling up their business. Since firms do not face a borrowing constraint, the accumulation of internal funds is not critical in the model either.

[32] If startups pay less than $\bar{\pi}_{\text{vc}}$, VCs will find other startups that are willing to accept slightly higher compensation. If VCs demand more than $\bar{\pi}_{\text{vc}}$, startups will find other VCs who are willing to accept slightly lower compensation. The stable match could be microfounded as an equilibrium under a proper market mechanism, such as a deferred acceptance algorithm.

The equity share of investors follows equation (7). The maximization problem above does not involve the equity share of investors. Appendix B.1 shows that this formulation is equivalent to maximizing the founder's value, which is given by the equity share of the founder multiplied by the firm's post-money valuation. The compensation for VCs is reflected in the extra equity stakes, as $\bar{\pi}_{\text{vc}}$ increases the amount of money raised ($-d_t$) in equation (10), resulting in higher equity stakes in equation (7).

3.3 Bank Financing

Bank-backed firms raise defaultable short-term debt instead of raising equity. Bank financing does not entail fixed costs ($c_{e,0}$) or flotation costs ($(-d)c_{e,1}$), as in equity financing. However, if a firm cannot pay back debt, it goes bankrupt. This procedure incurs default costs, and banks seize only a part of the firm's assets. Banks take into account the possibility of a firm's bankruptcy and the associated default costs and charge a bank loan rate such that the expected return of the loan is equal to the risk-free rate.

A firm raises debt b_{t+1} with price $q_t(b_{t+1}, k_{t+1}, I_t; n_t, \epsilon_t, o_t, \text{fail}_t, \hat{\theta})$ at time t and pays back b_{t+1} in the next period. The price of debt satisfies $q_t < 1$ and is a function of control variables (b_{t+1}: size of debt, k_{t+1}: capital input, I_t: innovation expenditures) and the firm's state variables. The bank loan rate is given by the inverse of the price $1/q_t$ and is equal to the risk-free rate, $1 + r$, if the default probability in the next period is zero.

In bank financing, the firm's cash flow is revenue minus the debt repayment:

$$o_t = y_t - b_t. \tag{11}$$

If the cash flow is negative, the firm either defaults on debt, in which case the firm loses all revenue and shuts down its business (i.e., Chapter 7 bankruptcy), or borrows additional money and rolls over the debt. The decision to default, denoted by $\Delta_{t+1} = 1$, is given by:

$$\Delta_{t+1} = \mathbf{1}\left\{o_{t+1} < 0 \text{ and } (V_{t+1} < 0 \text{ or exogenous exit shock})\right\}. \tag{12}$$

The firm chooses to default on the debt if the outstanding debt exceeds revenue and the firm does not operate in the next period, either because of the firm's negative present value or because of an exogenous exit shock.

Banks operate in a competitive market and charge a bank loan rate that satisfies the

following zero-expected-profit condition:

$$q_t(\cdot) = \frac{1}{1+r}\mathbb{E}_t\left[\underbrace{\mathbf{1}\left\{\Delta_{t+1}=0\right\}}_{\text{no default}} + \underbrace{\mathbf{1}\left\{\Delta_{t+1}=1\right\}}_{\text{default}}\left(\frac{y_{t+1}}{b_{t+1}} - c_d\right)\right]. \tag{13}$$

If a firm does not default on debt ($\Delta_{t+1}=0$), a bank receives full repayment on debt. In case of default ($\Delta_{t+1}=1$), a return on debt for the bank is given by $\left(\frac{y_{t+1}}{b_{t+1}} - c_d\right)$. This is because the firm transfers all revenue (y_{t+1}) to the bank upon default, but the default incurs costs that are proportional to the size of the debt.

A value function of a bank-backed firm is now formulated. At the beginning of each period, firms choose to continue their business, pay back their debt and exit, or default on the loan and exit. Thus, the firm's value is the maximum among the continuation value, cash flow, and the value of default, which is zero from the business owner's perspective.

$$V_t^{\text{Bank}}(n_t, \epsilon_t, o_t, \text{fail}_t; \hat{\theta}) = \max\{\underbrace{V_t^{\text{Bank},C}(n_t, \epsilon_t, o_t, \text{fail}_t; \hat{\theta})}_{\text{continue}}, \underbrace{o_t}_{\text{exit}}, \underbrace{0}_{\text{default}}\}.$$

The continuation value is given by the following:

$$\begin{aligned} V_t^{\text{Bank},C}(n_t, \epsilon_t, o_t, \text{fail}_t; \hat{\theta}) = & \max_{d_t \geq 0, \ell_{t+1}, k_{t+1}, I_t, b_{t+1}} d_t \\ & + \frac{1}{1+r}\left\{(1-\varphi)\underbrace{\mathbb{E}\left[V_{t+1}(n_{t+1}, \epsilon_{t+1}, o_{t+1}, \text{fail}_{t+1}; \hat{\theta})\right]}_{\text{no exit shock}} + \varphi \underbrace{\mathbb{E}\left[\max\left\{0, o_{t+1}\right\}\right]}_{\text{exogenous exit}}\right\} \\ \text{s.t. } & d_t + \underbrace{Rk_{t+1} + w\ell_{t+1} + I_t + c_f}_{\text{expenditures}} = \underbrace{(\hat{\theta}_{n_t} + \epsilon_t)^{1-\eta}(k_t^{\alpha}\ell_t^{1-\alpha})^{\eta} - b_t}_{=:o_t \text{(cash flow)}} + \underbrace{q_t(b_{t+1}, k_{t+1}, I_t)b_{t+1}}_{\text{new debt}}, \end{aligned} \tag{14}$$

where the price of debt q_t and the default decision Δ_{t+1} are given by equations (12) and (13), respectively. The probability of moving up the ladder ς_{t+1} is determined by innovation functions (4) and (5). If the cash flow is larger than total expenditures, the firm distributes the business surplus as a dividend. Otherwise, firms borrow money from a bank or cut expenditures to meet their budget constraints. The optimality conditions and numerical algorithms to compute the value function iteration are described in Appendix B.2.3.

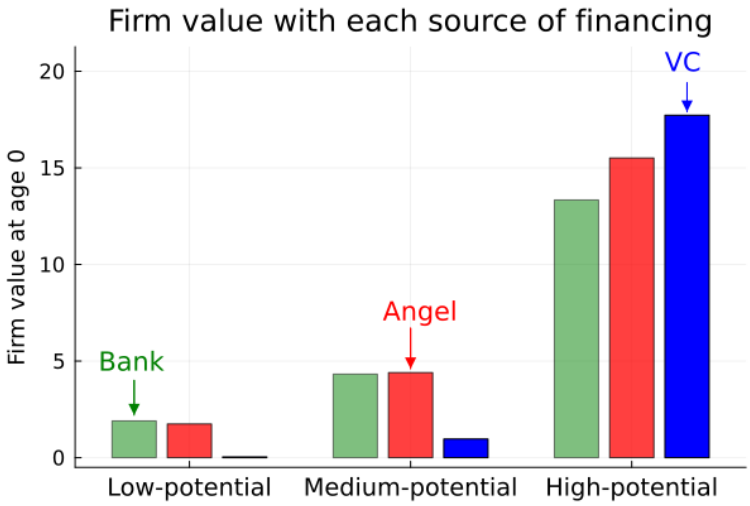


Figure 5: The Choice of Financing at Age 0

3.4 Choice of Financing

Firms with heterogeneous initial characteristics choose the source of financing that maximizes the firm's present discounted value:[33]

$$V_0(n_0, \epsilon_0, o_0, \hat{\theta}) = \max \left\{ V_0^{\text{VC}}(n_0, \epsilon_0, o_0, \hat{\theta}), V_0^{\text{Angel}}(n_0, \epsilon_0, o_0, \hat{\theta}), V_0^{\text{Bank}}(n_0, \epsilon_0, o_0, \hat{\theta}) \right\}.$$

Figure 5 shows an example of the firm's value in the model depending on growth potential. In the figure, high-potential firms attain the highest value if they choose VC financing. In contrast, low-potential firms obtain the highest value with bank financing. Intuitively, high-potential firms benefit the most from VCs' advice since the gain from successful innovation is exceptionally high.[34] Low-potential firms do not heavily invest in innovation, so the probability of default and the associated default costs in bank financing are low, as shown in Figure 14 in Appendix B.3.1. Medium-potential firms choose Angel financing since equity financing is more suitable than bank financing to finance innovation, as failure in innovation may lead to costly bankruptcy in bank financing, but the compensation for VCs is too expensive. This results in a monotonic relationship between a firm's potential and the mode of financing.

Firms are born with three initial characteristics $(\hat{\theta}, \epsilon_0, o_0)$. The growth potential $\hat{\theta} = \{\hat{\theta}_n\}_{n=1,\cdots,N}$ is summarized by a single number $\hat{\theta}_N$, which is the productivity at the last step of the ladder. Productivity levels in other rungs of the ladder $\{\hat{\theta}_n\}_{n=1,\cdots,N}$ are deter-

[33] Since the firm's value already takes into account the payment to VCs, it can be interpreted as the firm value from the owner's perspective.

[34] The model exhibits complementarity between a firm's potential and VCs' advice, as productivity increase due to successful innovation is larger for high-potential firms than low-potential firms. Fox et al. (2012) estimate a two-sided matching game between entrepreneurs and VCs and show complementarities.

mined by the following parametric function of $\hat{\theta}_N$ and n:

$$\hat{\theta}_n = \hat{\theta}_N \left[(1-\xi) \left(\frac{n-1}{N-1} \right) + \xi \left(\frac{n-1}{N-1} \right)^3 \right]. \tag{15}$$

This functional form satisfies $\hat{\theta}_{n=1} = 0$ and $\hat{\theta}_{n=N} = \hat{\theta}_N$. If $\xi = 0$, the firm's productivity increases linearly along the ladder. If $\xi > 0$, the ladder has a convex shape. The curvature of the ladder ξ is calibrated in Section 4.1 to match average firm growth in the data.[35] The growth potential $\hat{\theta}_N$ follows a log-normal distribution:[36]

$$\hat{\theta}_N \sim \text{LogNormal}(\mu_\theta, \sigma_\theta). \tag{16}$$

The initial exogenous productivity ϵ_0 also follows a log-normal distribution:[37]

$$\epsilon_0 \sim \text{LogNormal}(\mu_{\epsilon 0}, \sigma_{\epsilon 0}). \tag{17}$$

The initial assets o_0 are assumed to be the same for all firms.[38]

Free Entry Condition Entrants are ex-ante identical. Their growth potential $\hat{\theta}$ and exogenous productivity ϵ_0 are observed upon entry. The free entry condition ensures that the ex-ante value of the entry is equal to the cost of entry denoted by f^e:

$$\text{Value of Entry} := \sum_{\epsilon_0, \hat{\theta}} \underbrace{V_0(n_0, \epsilon_0, o_0, \text{fail}_0 = 0; \hat{\theta})}_{\text{firm value given an initial state}} \Phi(\epsilon_0, \hat{\theta}) = \underbrace{f^e}_{\text{cost of entry}}, \tag{18}$$

where the growth potential $\hat{\theta}$ and initial exogenous productivity ϵ_0 follow distributions (16) and (17). $\Phi(\epsilon_0, \hat{\theta})$ denotes the probability mass of firms with initial states $(\epsilon_0, \hat{\theta})$. All firms start from an initial step of the ladder ($n_0 = 1$) and with initial assets o_0. Since newborn firms have not failed in innovation, their failure status starts with $\text{fail}_0 = 0$

3.5 Aggregation

This subsection describes a representative household and capital renters. The economy is closed with market clearing conditions.

[35] The cubic term is introduced to allow for a steep, non-linear shape of the ladder. The quadratic term is dropped to reduce the number of parameters in calibration and to avoid a colinearity issue.

[36] To numerically solve the model, I draw nine values of $\hat{\theta}_N$'s from the log-normal distribution at the percentiles: [0.2, 0.5, 0.7, 0.85, 0.935, 0.9725, 0.9775, 0.989, 0.999]. The probability measures assigned to each percentile are $\frac{1}{100} \times$[40, 20, 20, 10, 7, 0.5, 0.5, 1.8, 0.2], respectively.

[37] The AR(1) process of ϵ in equation (3) is approximated by the Tauchen method. Appendix B.3.2 describes how grid points are defined. The number of grid points for ϵ is set to five.

[38] In practice, initial assets may be correlated with the growth potential $\hat{\theta}_N$ and initial productivity ϵ.

Representative Household The representative household supplies a fixed amount of labor $\bar{L}$ every period. All profits from heterogeneous firms and venture capitalists are returned to the household, denoted by Π_t.[39] The infinitely lived household maximizes the lifetime utility subject to a budget constraint:

$$\max_{c_t, a_{t+1}} \sum_{t=0}^{\infty} \beta^t u(c_t) \text{ s.t. } c_t + a_{t+1} = w_t \bar{L} + (1+r)a_t + \Pi_t,$$

where profits Π_t are given by $\sum_i [\pi_{it} + \mathbf{1}\{\text{VC}_i\}\bar{\pi}_{\text{vc}}]$. Each firm is indexed by i. The household's saving a_t returns a risk-free rate, $1+r$. Flow utility is assumed to be CRRA (constant relative risk aversion). The household chooses consumption and savings, following the Euler equation:

$$u'(c_t) = \beta(1+r)u'(c_{t+1}).$$

In the steady state, the Euler equation implies:

$$1 + r = \frac{1}{\beta}. \tag{19}$$

Hence, the steady-state risk-free rate is given by the inverse of the discount factor β.

Capital Renters Capital renters collect savings from the representative household and rent capital to firms with a rental rate R. Since capital depreciates with rate δ, the rental rate of capital R is given by:[40]

$$R = r + \delta. \tag{20}$$

The capital rental market is competitive, and capital renters make zero profits.

Market Clearing Conditions The labor market, capital market, and final goods market clear in equilibrium. First, the aggregate labor demand by heterogeneous firms is equal to the exogenous labor supply of the household:

$$\sum_i \ell_{it} = \bar{L}. \tag{21}$$

However, initial assets at the firm level are typically not observed in the data.

[39] Households also pay for initial assets of newborn firms and receive cash flows of firms hit by an exogenous exit shock. Banks and Angel investors generate zero profits.

[40] The maximization problem of capital renters can be formulated as $\max_{K_t} (R+1-\delta)K_t - (1+r)K_t$.

Second, aggregate capital rented to heterogeneous firms by capital owners is equal to the household's savings:

$$\sum_i k_{i,t+1} = a_{t+1}.$$

Finally, the total output produced plus undepreciated capital is equal to the sum of all expenditures, household consumption, and capital in the next period:

$$Y_t + (1-\delta)K_t = \underbrace{\sum_i [I_{it} + c_f + f^e \mathbf{1}\{\text{age}_i = 0\} + \mathbf{1}\{\text{equity}_i\}(c_{e,0} + c_{e,1}(-d_{it})) + \Delta_{it} c_d b_{it}]}_{\text{all expenditures}(=:\text{EXPND}_t)} + C_t + K_{t+1},$$

where $Y_t := \sum_i y_{it}$ and $K_t := \sum_i k_{it}$ denote the aggregate output and capital, repspectively. In the steady state, capital in the economy is invariant over time. Therefore, the household's consumption in the steady state is given by:

$$C = Y - \text{EXPND} - \delta K.$$

The distribution of heterogeneous firms with state variables $(n_i, \epsilon_i, o_i, \text{fail}_i; \hat{\theta}_i)$ is stationary in the steady state.[41] Wages are set to clear the free-entry condition (18). In the steady state, the rental price of capital R is determined by the discount rate of the household and the depreciation rate, as in equations (19) and (20). The price of final goods is normalized to one. The measure of entrants in the steady state is given by M^e.

4 Quantitative Analysis

The model is calibrated to capture the salient features of firm dynamics and explains the substantial growth and upfront investments of VC-backed firms as untargeted moments. The calibrated model is used to quantify the importance of VC financing at the micro and macro levels.

4.1 Calibration

The model features 23 parameters listed in Table 6. Nine parameters are fixed exogenously. The remaining 14 parameters are internally calibrated to minimize the distance between model-implied moments and targeted data moments. The targeted moments

[41] Appendix B.3.3 describes how the stationary distribution is computed numerically.

Table 6: Parameters in the Model

Parameters	Values	Description	Identification
Innovation Function			
γ_0	5.61	intercept	Pr(2nd VC round)
γ_1	0.46	coef on $\log(I)$	equity share of VC (age 0)
γ_2	-0.96	coef on $\log(1+\hat{\theta}_N)$	equity share of Angel (age 0)
γ_3	-4.47	coef on failure	average firm size over life cycle
γ_4	0.12	coef on VC	VCs' extra equity share (3%)
γ_5	0.08	coef on $\log(n)$	Pr(5th VC round)
Productivity Distribution			
μ_θ	-4.09	mean of $\log(\hat{\theta}_N)$	fraction of Gazelles
σ_θ	3.81	std of $\log(\hat{\theta}_N)$	Pareto tail
μ_{ϵ_0}	1.68	mean of $\log(\epsilon_0)$	average firm size over life cycle
σ_{ϵ_0}	4.01	std of $\log(\epsilon_0)$	average firm size over life cycle
ρ	0.8	persistence	exogenous
σ_ν	1	std of AR(1) shocks	exogenous
Financial Frictions			
$c_{e,0}$	0.29	fixed cost of equity	top 2% choose equity
$c_{e,1}$	0.04	unit cost of equity	exogenous
c_d	0.41	default costs	loan recovery rate
o_0	0.41	initial assets	wage of one worker
Other Parameters			
r	0.08	risk-free rate	standard value (2yr)
δ	0.2	capital depreciation	standard value (2yr)
α	0.33	capital share	standard value
η	0.85	span of control	standard value
φ	0.08	exit shock	exogenous (2yr)
ξ	0.69	convex ladder	average firm size over life cycle
c_f	0.36	operating costs	exit rate

are shown in Figure 6 and Table 7.

Exogenous Parameters The model period is two years, so the risk-free rate is set to $r = 0.08$. Capital depreciates at rate $\delta = 0.2$ biannually. The coefficient on capital in the production function (1) is $\alpha = 0.33$. The span-of-control parameter is $\eta = 0.85$ (e.g., Midrigan and Xu, 2014). An exogenous biannual exit rate is set to $\varphi = 0.08$.[42] The persistence parameter in the exogenous productivity process (3) is chosen to be $\rho = 0.8$.[43] The standard deviation of exogenous productivity shocks σ_ν is set to unity. The unit cost of raising capital is set equal to $c_{e,1} = 0.04$, within a range of values reported in the literature (e.g., Gomes, 2001 and Hennessy and Whited, 2007). Finally, startups launch their businesses with initial assets o_0, which are chosen to be equal to the wage of one worker.[44]

Fitted Parameters The rest of the parameters are chosen to match empirical moments. The first set of parameters $(\gamma_0, \gamma_1, \gamma_2, \gamma_3, \gamma_4, \gamma_5)$ in the innovation function (5) are unique to this study. Note that estimation of innovation functions is generally challenging as researchers do not directly observe innovation inputs and innovation outputs.[45] This study estimates the coefficients using the characteristics of VC deals and Angel deals observed in the data.

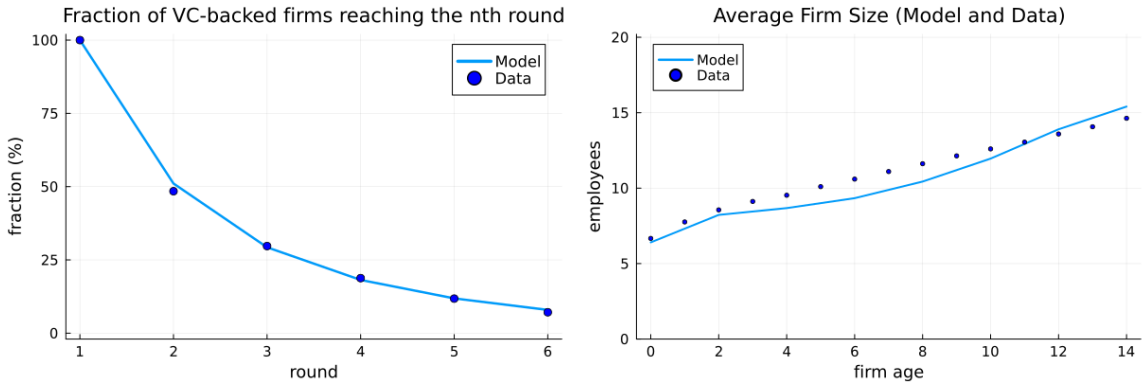
In practice, VC-backed firms raise money with multiple rounds, allowing venture capitalists to assess the firm's performance before injecting additional capital. This implies that raising additional VC rounds is an indication of satisfactory business performance. Hence, parameters are chosen such that the model-implied probabilities of raising the second and fifth rounds hit the empirical counterparts (48.4% and 11.8%, respectively). Figure 6 (left panel) shows that the model and the data match well in this dimension. This identifies the coefficients γ_0 (intercept) and γ_5 (the coefficient on $\log(n_t)$) in the innovation function, which governs the innovation success rates at the initial and upper

[42] Sterk et al. (2021) estimate an annual exogenous exit rate of around 4.1% in their calibration.

[43] Foster et al. (2008) report persistence of various TFP measures and demand shocks ranging from 0.76 to 0.97 at an annual rate. Instead of exogenously setting this parameter, one could internally calibrate ρ to be consistent with the autocorrelation of firm size in the data at the expense of computational burden. The calibrated model predicts the autocorrelation of log(employment) of around 0.8, within the range of biannual autocorrelation of around 0.75 to 0.9 at various ages, reported by Sterk et al. (2021).

[44] Robb and Robinson (2014) report that the average owner equity of startups is around $33,000 in 2004. The average wage in 2004 was around $36,000, according to the Social Security Administration.

[45] A typical approach would be to measure innovation inputs with R&D expenditure and innovation outputs with patent filing. However, most firms in the economy do not invest in R&D and do not produce patents. Hence, this approach would significantly narrow down the scope of innovation. R&D is defined as "creative and systematic work undertaken in order to increase the stock of knowledge and to devise new applications of available knowledge," according to the BRDIS survey form. It does not include activities related to the firm's business expansion, such as market research, management studies, and the creation of new software based on known methods and applications. My study examines firm growth not necessarily based on R&D activity or patent generation.



Note: The left panel shows the fraction of VC-backed firms that raised the n-th VC round ($n = 1, \cdots, 6$) in Crunchbase (1980–2022). The right panel shows the average firm size of all firms in the economy from age 0 to 14. The data are from Sterk et al. (2021)'s replication file.

Figure 6: Probability of Subsequent VC Deals and Average Firm Size

steps of the ladder conditional on innovation expenditures.

The parameter $\gamma_3 < 0$ measures the negative impact of innovation failure on future innovation. If the magnitude of γ_3 is large enough, VC-backed firms will not raise the next VC round in case of innovation failure. This parameter also affects the average growth of all firms, which helps pin down the value.

The elasticity of the probability of successful innovation to innovation expenditures (γ_1; the coefficient on $\log(I_t)$) is pinned down based on the equity stakes of VC investors at firm age 0 and the probability of raising the second VC round. The idea is that since VC-backed firms raise money to invest primarily in innovation and the investor's equity share is approximately equal to the money injected divided by the firm value, the equity share contains information about the size of innovation expenditures relative to firm value.[46]

The parameter γ_2 (the coefficient on $\log(1+\hat{\theta}_N)$) determines the relation between the firm's potential and innovation success rate. This parameter is pinned down by targeting the average equity stake of Angel investors in the first deal. The idea is that this moment

[46]The following derivations show the relation between equity shares of investors and the parameter γ_1. The first order condition (FOC) with respect to innovation expenditure I gives:

$$\frac{I}{\mathbb{E}\left[V_{t+1}(\text{success}) - V_{t+1}(\text{fail})\right]} = \frac{\gamma_1 \varsigma^*(1-\varsigma^*)}{1+r},$$

where the firm value is given by: $V_t = \frac{1}{1+r}\left\{\varsigma V_{t+1}(\text{success}) + (1-\varsigma)V_{t+1}(\text{fail})\right\}$. Under simplifying assumptions, there is a one-to-one mapping between the parameter γ_1 and the equity share of investors. If $V_{t+1}(\text{fail}) \approx 0$, the two equations imply:

$$\frac{I}{V_t} \approx \gamma_1(1-\varsigma^*). \tag{22}$$

In addition, if all raised money is used for innovation expenditure and if VCs make zero profits (i.e., $\frac{I}{V_t} \approx$ equity share), γ_1 can be derived in closed form, using the equity share of VC investors.

is informative about the intensity of innovation activities of Angel-backed firms.[47]

The value-adding effect of VCs, denoted by γ_4, is identified by the extra equity stake of venture capitalists documented in the empirical section. The idea is to measure the perceived value-adding effect of VCs from the startup founder's perspective. The parameter γ_4 is jointly determined with the VCs' management fees $\bar{\pi}_{\text{vc}}$. Firms at the 99.8th percentile (top 0.2%) are at the threshold and indifferent between VC financing and Angel financing. While VC financing boosts the probability of successful innovation, VCs require 3.3% of extra equity stake. The parameter γ_4 and VCs' management fees $\bar{\pi}_{\text{vc}}$ are set such that the benefit and cost balance out for firms at the top 0.2% threshold.

The second set of parameters $(\mu_\theta, \sigma_\theta, \mu_{\epsilon_0}, \sigma_{\epsilon_0})$ govern the productivity distribution. The distribution of growth potential $\hat{\theta}_N$, given by LogNormal$(\mu_\theta, \sigma_\theta)$, is set to match the fraction of Gazelles among all startups, 0.87%, taken from Table 2, and the Pareto tail of the firm-size distribution, 1.06, among firms with more than 100 employees, as reported by Luttmer (2010). The distribution of initial productivity ϵ_0, given by LogNormal$(\mu_{\epsilon_0}, \sigma_{\epsilon_0})$, regulates the average firm size in early ages.

The curvature of the productivity ladders (ξ) and wage (w) also govern the evolution of average firm size. Higher wage w lowers the average firm size, while the curvature of the ladder ξ forms the trajectory of average firm size growth. The average size of all firms from age 0 to 14 is targeted, as depicted in the right panel of Figure 6. The model fits the average firm size over the life cycle reasonably well. The coefficient c_f is chosen to match the exit rate, as higher operating costs c_f induce more firms to exit.[48]

Finally, the two parameters $(c_{e,0}, c_d)$ related to financial frictions are set such that 2% of all firms choose equity financing (VC or Angel) and that banks recover 54% of defaulting loans in case of bankruptcy, a value taken from Ottonello and Winberry (2020) and Gomes and Sarkisyan (2023).[49] Parameter $c_{e,0} = 0.29$ implies that equity-backed firms pay fixed costs each period that are less than hiring one additional worker. Parameter $c_d = 0.41$ implies that defaulting on bank loans incurs costs equivalent to 41% of outstanding debt.

Overall, 19 moments are targeted to set 14 parameter values and two endogenous

[47] Based on the approximated relation (22), one can infer that a lower equity share is related to a higher innovation success rate ς.

[48] It turns out that the high exit rate observed in the data is difficult to explain with the model. A higher exit rate implies high operating costs. However, this would give rise to higher growth of surviving firms and would not be in line with the average firm size in the data. One can see that Sterk et al. (2021) also struggle to explain the average firm size and exit rate jointly. A potential solution could be to introduce size-dependent operating cost shocks, as in Kochen (2022).

[49] Sohl (2012) reports that the Angel market invested in over 10 times more deals than the VC market in the 2000s. For example, Angel investors invested in more than 57,000 startups in 2009. In the Kauffman Survey (2004), the fraction of firms that raise VC or Angel financing at age 0 is around 3.2%. In the Survey of Business Owners (2007), the fraction of equity-backed firms is around 1%. Since the precise fraction of Angel-backed firms in the economy is unknown, 2% of all newborn firms are assumed to raise equity financing in this study.

Table 7: Remaining Target Moments

Description	Target	Model
Fraction of "Gazelles"	0.87%	0.92%
Pareto tail	1.06	1.10
Average equity share at age 0 (VC)	33.0%	34.2%
Average equity share at age 0 (Angel)	23.2%	22.6%
Exit rate (annual) at age 0-20	9.4%	8.0%
Loan recovery rate	54%	54%
Fraction of VC-backed firms	0.2%	0.2%
Fraction of equity-backed firms	2%	2%
Extra equity share of VC investors	3.3%	3.3%

*N*ote: The data moments on the fraction of Gazelles, the fraction of VC-backed firms, and the extra equity share of VC investors are taken from the descriptive evidence in Section 2. Gazelles are firms with an average annual growth rate above 20% in the first five years and employment of 100 or above by age 20. The Pareto tail of firm size distribution, the average equity share of VCs and Angel investors, exit rate, and loan recovery rate are taken from Luttmer (2010), Pitchbook (1980-2022), BDS (2000-2019), and Ottonello and Winberry (2020), respectively. The fraction of equity-backed firms is set to be ten times the fraction of VC-backed firms.

prices (w and $\bar{\pi}_{\text{vc}}$) in the model.[50] The calibrated parameters minimize the sum of squared distances between targets and model-implied moments.[51]

4.2 Model Fit

The calibrated model explains the substantial growth and upfront investments of VC-backed firms documented in Section 2. Furthermore, the model's prediction on the evolution of equity shares and the average total money raised in VC financing and Angel financing are compared with the data. All those moments are untargeted and match their empirical counterparts reasonably well.

Two Motivational Facts The first motivational fact is the substantial growth of VC-backed firms compared to non-VC-backed firms in the sample of Gazelles. To compare the model and the data, I take the simulated sample of Gazelles in the model following the same definition as in the data and compute the average firm size at each age,

[50] I take two moments in the left panel of Figure 6, eight moments in the right panel of Figure 6, as firm size at age (0,2,4,6,8,10,12,14) are targeted, and nine moments in Table 7. The prices in the steady state are $w = 0.42$ and $\bar{\pi}_{\text{vc}} = 7.2$. In the baseline economy, the measure of entrants M^e is normalized to one. The calibrated model implies that the value of entry is 2.5.

[51] The Nelder-Mead method is used in calibration. The eight moments regarding the average firm size are weighted by 5/8, as the eight moments are primarily used to identify four parameters and one price (w). The moment about the exit rate and the Pareto tail are weighted by 1/5 and 2, respectively, given the importance of the moments in the quantitative model. Fitting the exit rate turns out to be challenging, as discussed in footnote 48, and is not the main focus of this paper, while the Pareto tail is a critical moment for the size of the largest firms in the economy. The other moments have the weight of one.

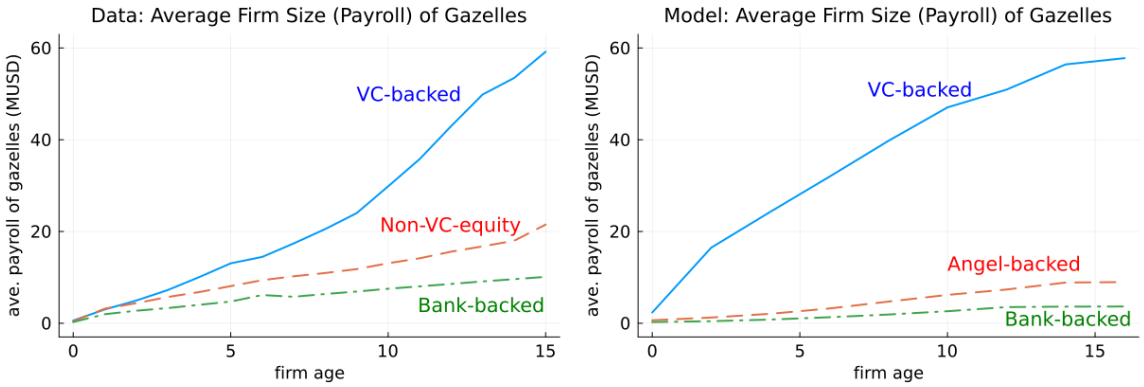
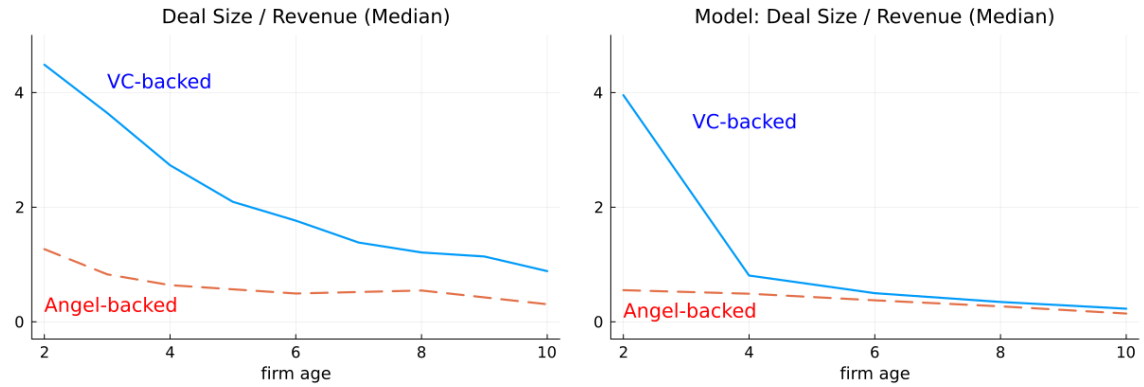
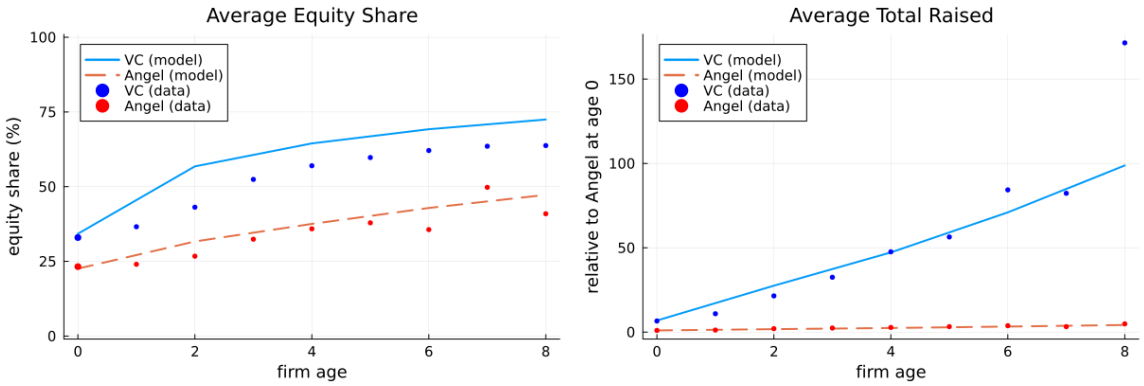


Figure 7: Growth Trajectory of Gazelle Firms (Data vs. Model)



Note: The figures on the left show the motivational facts documented in Section 2. The right panel shows the model's predictions. Since firms earn zero revenue at age 0 in the model, deal size over revenue at age 0 is not depicted in the right panel.

Figure 8: Upfront Investment of High-Potential Firms (Data vs. Model)



Note: The average equity share at age 0 is targeted in calibration, while other moments are untargeted. Equity share represents the total equity share of investors. The total money raised is shown relative to the average money raised at age 0 in Angel financing. The data are drawn from Pitchbook (1980-2022).

Figure 9: Average Equity Shares and Total Money Raised (Data vs. Model)

separately for VC-backed, Angel-backed, and bank-backed firms. Figure 7 demonstrates the average growth trajectories over the life cycle in the data (left panel) and in the model (right panel). The model is able to explain the monotonic relationship between the source of financing and firm growth.[52]

The second motivational fact is the upfront investment of equity-backed firms. To assess the model's prediction, I take the simulated sample of VC-backed and Angel-backed firms and compute the ratios of the amount of money raised (i.e., deal size) to the firm's revenue. The median values over the firm's life cycle are depicted in Figure 8. As in the data, VC-backed firms make large upfront investments, several times larger than their revenue, in their early life.

Other Untargeted Moments In addition to the two results above, the model's predictions are compared to other dimensions of the data. Figure 9 compares the model's prediction with the data on the evolution of the average equity share of investors and the total money raised for VC-backed and Angel-backed firms over their life cycle.[53] According to the data, VC-backed firms typically transfer more than half of their ownership stake to their investing partners after several rounds are raised. Angel-backed firms also offer additional ownership stakes over the course of financing but less so than VC-backed firms. These features are well-captured in the model (left panel). In the right panel, the

[52]In calibration, I target the average firm size of all firms and the Pareto tail of the stationary firm-size distribution, neither of which directly controls the growth trajectory of Gazelles in the three groups. In Figure 7, firm size is measured in payroll in the data. In the model, I first compute the average firm size in terms of employment. Then, I multiply the values with $40,000 to make a direct comparison with the data.

[53]The data are drawn from Pitchbook, and firms are observed only in the year the firm obtained new external funding. To construct the model's counterpart, I take an average of cumulative equity shares of investors and the total money raised in the sample of firms with positive external funding. In the data, I drop firms that take the first financing deal after age 2. In the model, all firms raise their first financing deal at age 0. Only a small fraction of firms above age 8 are observed in the data.

total money raised from age 0 to 8 is reported. The variables are normalized by the funding size of Angel-backed firms at age 0. In the data, the amount of money raised by VC-backed firms is significantly larger than Angel-backed firms and increases substantially over the life cycle.[54] This aspect is also reasonably well-captured in the model. Finally, the annualized default rate among all firms in the model is around 0.5% and is close to a value reported in the literature.[55]

4.3 Counterfactual Experiments

This study quantifies the role of venture capitalists at the micro and macro levels. In the counterfactual experiment at the micro level, the hypothetical growth path of VC-backed firms is simulated while the wage is kept unchanged. In a counterfactual economy that differs from the baseline economy in available sources of financing, the wage is determined in equilibrium such that the free entry condition (18) is satisfied. In the steady state, the measure of entrants M^e is constant over time and is determined by the labor market clearing condition (21), which can be written as:

$$M^e \sum_{h=0}^{\infty} \sum_{\epsilon_0, \hat{\theta}} \ell_h(n_h, \epsilon_h, o_h; \hat{\theta}) \, \Phi(\epsilon_0, \hat{\theta}) = \underbrace{\bar{L}}_{\text{exogenous labor supply}} .$$

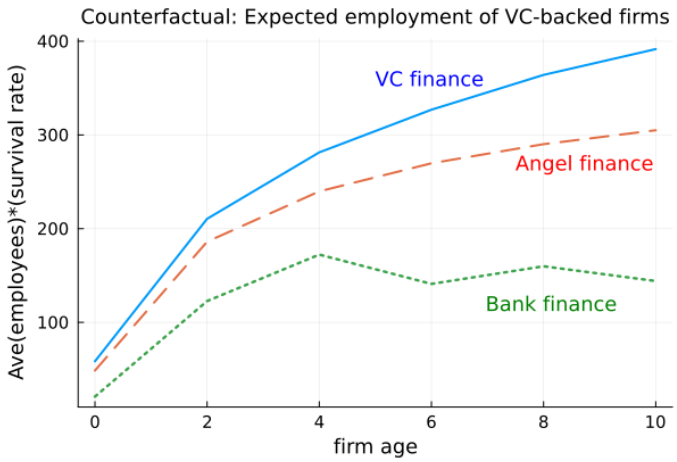
The total labor demand is the sum of labor demand by each cohort $h = 0, 1, \cdots$. Firms in cohort $h = 0$ are age 0 in the economy. Firms with $h = 1$ were born one period before and are at age 2, since one period in the model is two years.

The role of VCs The contribution of VCs' advice at the micro level is quantified by predicting the growth trajectories of VC-backed firms if they instead raise funding from Angel investors or banks. Figure 10 shows the expected growth trajectories of VC-backed firms under alternative sources of financing, which is computed by multiplying the average employment of surviving firms with the survival rate. The counterfactual experiment predicts that firms born with the same growth potential would grow 22% more in expectation with VC financing at age 10 compared to the expected firm size with Angel financing.[56] Conversely, the expected size of these firms would decline by 53% with bank financing relative to Angel financing at age 10, demonstrating the importance of equity

[54] In the data, Angel-backed firms raise an average of $1.2 million at age 0 and $4 million by age 8. VC-backed firms initially raise $5.6 million at age 0 and $103 million by age 8 on average.

[55] Gomes and Sarkisyan (2023) report a corporate default rate of 0.4% from Moody's. According to the Administrative Office of the US Courts, there were 24,735 business bankruptcy filings in 2015, which is around 0.4% of all firms. Source: `https://www.uscourts.gov/`.

[56] In comparison to the literature, Sørensen (2007) estimates a two-sided matching model with heterogeneous firms and VCs and concludes that endogenous sorting is almost twice as important as the influence of VCs on the firm's performance.



*N*ote: The expected employment is defined as the average employment of surviving firms multiplied by the survival rate. The solid line shows the expected employment of VC-backed firms simulated with the calibrated model. The expected employment of the same firms with alternative sources of financing is shown in the dashed line(Angel financing) and in the dotted line (bank financing).

Figure 10: Prediction on the Growth Trajectory of VC-backed Firms with Alternative Sources of Financing

Table 8: Counterfactual Economies

	Without VCs (%)	Without any equity (%)	More VCs (%)
consumption (EV)	–0.43	–0.76	0.15
output	–0.16	–0.50	0.10
wage	–0.18	–0.43	0.04
emp share of gazells	–5.71	–6.32	2.14

*N*ote: This table shows percentage changes of aggregate variables in the steady-state analysis from the baseline economy to counterfactual economies without VC, without any equity financing, and with more VCs. Without the VC sector, all firms raise either Angel or bank financing. Without equity financing, all firms rely on bank financing. An economy with more VCs has twice as many VCs as the baseline economy. The percentage change in consumption is identical to the equivalent variation (EV), as the household's utility depends only on consumption. The equilibrium wage clears the free entry condition.

financing for high-growth firms.

A counterfactual exercise at the macro level considers an economy in which VC financing is unavailable. VC-backed firms in the status quo would instead raise Angel financing in this scenario. Table 8 shows that aggregate output in the steady state is predicted to decline by around 0.2%. Aggregate consumption declines by around 0.4%, which is equivalent to around 70 billion dollars based on the total consumption expenditures in the US in 2022. Hence, advice from VCs has a sizable impact on the economy.[57] In the counterfactual economy, wages decline such that the value of entry is kept unchanged. The employment share of Gazelles also declines since high-potential firms are less likely to succeed in innovation.

The second column shows the simulation results with an economy without any equity (VC or Angel) financing. This leads to around 0.8% lower aggregate consumption in the steady state. Equity-backed firms in the status quo now rely on bank financing and are less likely to seize their growth opportunities, leading to a loss in aggregate consumption. Finally, the third column reports the simulation results in an economy with more VCs, where the fraction of VC-backed firms is 0.4% instead of 0.2%.[58] If the number of VCs in the economy doubles, more startups benefit from VCs' advice. In addition, more VCs in the economy decreases the equilibrium management fees of VCs, further facilitating the growth of VC-backed firms.

5 Concluding Remarks

This paper quantitatively examines the high growth of VC-backed firms in the US economy. Empirically, an administrative dataset is used to document the substantial growth of VC-backed firms at the micro level and their contribution to growth at the macro level. VC-backed firms are characterized by their large upfront investments and extra equity stakes acquired by VC investors. Based on the evidence, this paper develops a firm dynamics model over the life cycle with innovation-based endogenous firm productivity

[57] Personal consumption expenditures in US GDP in 2022 are around 17.5 trillion dollars (Source: US Bureau of Economic Analysis). As a reference, the US GDP declined by around 2.6% in 2009 during the Great Recession (Source: FRED). Lucas (1987)'s calculation of the cost of the business cycle is less than 0.1% of consumption, which is considered a small value. The value of 0.4% in this study is somewhat in between, so I interpret it as a medium-scale impact. Note that the model does not consider the positive spillover of innovation or its impact on economic growth; thus, the prediction can be seen as a lower bound of the role of VCs in the aggregate economy. In addition, the total labor supply is assumed to be constant regardless of changes in wages.

[58] A baseline economy is recomputed with ten values of growth potential ($\hat{\theta}_N$) drawn from the log-normal distribution at the percentiles [0.2, 0.5, 0.7, 0.85, 0.935, 0.9725, 0.9775, 0.988, 0.997, 0.999] and the probability measures assigned to each percentile $\frac{1}{100} \times$[40, 20, 20, 10, 7, 0.5, 0.5, 1.6, 0.2, 0.2], respectively. Firms in the last two percentiles raise VC financing in the counterfactual economy. The management fees of VCs ($\bar{\pi}_{\text{VC}}$) are computed such that firms at the 99.6 percentile in growth potential are indifferent between VC and Angel financing.

and the choice of financing from VC, Angel, and bank financing. The model explains the growth trajectories of firms with different financing sources and implies that VCs' managerial advice accounts for around one-fourth of the growth of VC-backed firms. The importance of VC financing in the macro economy is quantified in a general equilibrium analysis.

Based on the results of this paper, several directions may be pursued. First, firm growth is analyzed in a stationary equilibrium in this paper. It will be beneficial to extend the model to an endogenous growth model in which the economy sustains long-run growth. Then, the impact of the VC sector on aggregate growth can be analyzed.

Second, the two roles of venture capitalists examined in this paper are equity-based funding and managerial advice. Another role of VCs is to screen startups and monitor the usage of funds by VC-backed firms. An extension to incorporate uncertainty in a firm's potential and information asymmetry between firm owners and investors will help us understand the impact of information frictions on firm growth and aggregate output.

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A Appendix for the Empirical Results

A.1 Description of the Data

Longitudinal Business Database (LBD) The LBD contains records of all non-farm employer establishments that operate in the United States. The database is based on the administrative records from business tax filings that are shared by the Internal Revenue Service (IRS) with the US Census Bureau. The Census Bureau restores the longitudinal link of establishments across time based on tax identifiers and establishment names and addresses. See Chow et al. (2021) for the Census Bureau's recent redesign of the LBD.

A challenge is to construct longitudinally consistent firm identifiers. A firm is an economic unit consisting of single or multiple establishments under common ownership. The LBD historically used `firmid`, which is "0" followed by Employer Identification Number (`EIN`) for single-unit firms and `alpha` followed by "0000" for multi-unit firms. The `firmid` uniquely identifies all firms in each year, but the longitudinal linkage is not complete. This is because `firmid` may change over time for reasons unrelated to ownership changes.

I partially restore firm links over time for two cases, refering to Dent et al. (2018). First, if a firm transitions from a single-unit to a multi-unit firm (in this case, `firmid` always changes by construction) while preserving its `EIN`, then I treat the two firms with different `firmid` as the same firm. Second, a firm obtains a new `EIN` when its legal form of organization (LFO) transforms. Therefore, if a firm changes its LFO while preserving its `lbdnum`, a longitudinal establishment identifier created in the Census Bureau, and single-unit status, I treat the two firms as the same firm. These two efforts are particularly important for keeping track of high-growth firms and VC-backed firms as they are likely to change their single-unit/multi-unit status and LFO over the life cycle. These efforts also reduce the number of spurious firm births and deaths.

Survey of Business Owners (2007) The Survey of Business Owners (SBO), conducted by the Census Bureau, asks a large sample of firms in the US about their business and owner characteristics. Around 62% of the 2.3 million businesses in the survey in 2007 responded. The fraction of each group of firms is tabulated in Table 9, using sampling weights in the dataset.

The businesses in the SBO (2007) that responded to the questionnaire about the sources of capital are linked to the LBD. VC-backed firms are identified based on the name and address matching with the external VC datasets. Among the firms that received an early-stage investment in exchange for their equity, according to the questionnaire in the SBO, those that did not receive VC investment are regarded as non-VC-equity-

Table 9: Firms in SBO (2007)

	Fraction in the economy (%)	Odds of being Gazelles (%)
VC-backed firms	0.20	46.5
Non-VC equity	0.75	10.8
Bank-financed	31	2.57

*N*ote: The sample consists of businesses in the SBO (2007) that replied to the questionnaire about the sources of capital and are linked to the LBD. Gazelles are defined based on their payroll and are firms with an average annual growth rate of payroll above 20% in the first five years and whose payroll reaches 100 or above multiplied by the average payroll in the economy by age 20. Observations are weighted by the sampling weights in the SBO. FSRDC Project Number 2633 (CBDRB-FY23-P2633-R10186).

financed firms. Bank-backed firms are those receiving a "business loan from a bank or financial institution" or "government-guaranteed business loan from a bank or financial institution" but not obtaining VC financing or non-VC equity financing.[59]

Table 9 reports the fraction of payroll Gazelles in the three groups and shows that VC-backed firms are more likely to be Gazelles (46.5%) than non-VC-equity-backed firms (10.8%) and bank-backed firms (2.57%). Unlike the fraction of Gazelles among newborn firms documented in Table 2, this table shows the fraction of Gazelles among firms operated in 2007. The fraction of Gazelles among VC-backed firms in this sample (46.5%) is higher, as VC-backed firms that did not achieve high growth are more likely to have exited.

SIRD (1980-2007) and BRDIS (2008-2019) The Survey of Industrial Research and Development (SIRD) from 1980 to 2007 and the Business R&D and Innovation Survey (BRDIS) from 2008 to 2019 provide information on the R&D activity of US-based companies. The National Science Foundation (NSF) sponsors the surveys. Approximately 25,000–40,000 companies are surveyed each year. The survey's name changed in 2007, but there is no break in the series of items collected in both surveys. The key variable analyzed in this paper is the total costs incurred for R&D. I link the SIRD/BRDIS datasets to the LBD using census identifiers, referring to Cohen (2023).

VC datasets This study examines four VC datasets: Crunchbase, Pitchbook, Preqin, and VentureXpert. Crunchbase granted the author access to its database for research

[59]SBO (2007) asks about the sources of capital. The choices are as follows: Personal/family savings of owner(s), Personal/family assets other than savings of owner(s), Personal/family home equity loan, Personal/business credit card(s), Business loan from federal, state, or local government, Government-guaranteed business loan from a bank or financial institution, Business loan from a bank or financial institution, Business loan/investment from family/friend(s), Investment by venture capitalist(s) (an early-stage investment in exchange for ownership equity by an individual, outside group, or business not directly involved in the overall operation and management of the business), Grants, Other source(s) of capital, Don't know, and None needed.

purposes (URL: `www.crunchbase.com`). The other three VC datasets are accessed from the Wharton Research Data Services (WRDS) and the Penn Libraries.

I merge the Business Register with each of the four VC datasets via name and address matching. Before the matching procedure, all firm names and addresses in both the Business Register and VC datasets are cleaned using the Stata commands `stnd_compname` and `stnd_address`. For Crunchbase, Pitchbook, and VentureXpert, the matching procedure uses firm name, state, city, zip code, and street name. For Preqin, only firm name, state, and city are used. The matching is implemented sequentially in several passes, with each pass applying less stringent criteria than the previous one. This approach helps prevent valid matches from being excluded while maintaining a reasonable number of matches. To reduce false positives, firms matched to a VC dataset but presumably not VC-backed, based on the SDC New Issues and the IPO dataset compiled by Jay Ritter (URL: `site.warrington.ufl.edu/ritter/ipo-data/`), are dropped. Further details on the matching procedure are provided in the CES technical note (Ando and Liu, 2025).

A.2 Growth Accounting

This subsection describes the procedures of growth accounting, following Haltiwanger et al. (2013). An establishment-level DHS growth rate g_{it} is defined as:

$$g_{it} = \frac{E_{i,t} - E_{i,t-1}}{X_{it}}, \text{ where } X_{it} = \frac{1}{2}\left(E_{i,t-1} + E_{i,t}\right).$$

$E_{i,t}$ denotes employment or payroll in establishment i at time t.

A firm-level growth rate g_{ft} is a weighted sum of the growth rates of establishments that belong to firm f:

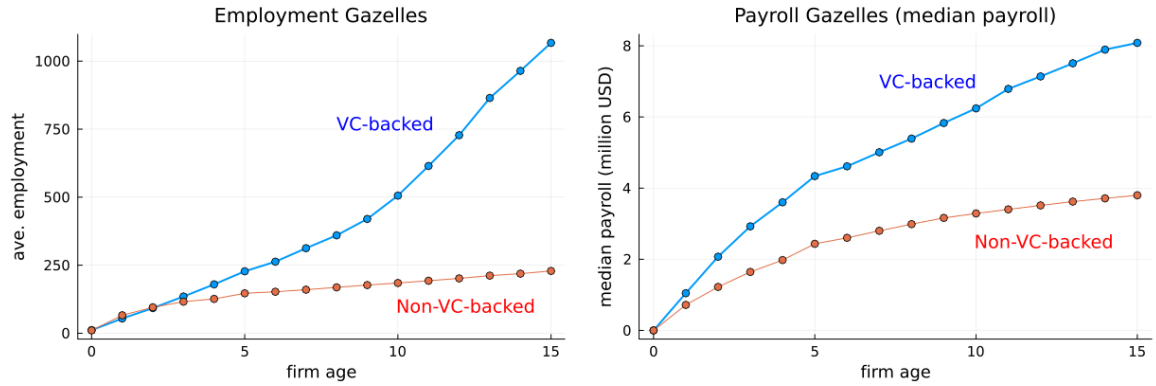
$$g_{ft} = \sum_{i \in f} \left(\frac{X_{it}}{X_{ft}}\right) g_{it}, \text{ where } X_{ft} = \sum_{i \in f} X_{it}.$$

The size of firm f is denoted by X_{ft}, and the size of establishment i that belongs to firm f is denoted by X_{it}.

Based on the DHS firm growth rates, the growth rate of a group of firms (g_{st}) is computed as a weighted sum of the firm growth rates within group s:

$$g_{st} = \sum_{f \in s} \left(\frac{X_{ft}}{X_{st}}\right) g_{ft}, \text{ where } X_{st} = \sum_{f \in s} X_{ft}.$$

The contribution of group s to aggregate growth in a particular year t is computed



Note: The sample consists of employment Gazelles in the left panel and payroll Gazelles in the right panel. The average employment and median payroll are reported separately for VC-backed firms and non-VC-backed firms. Data: LBD (1980-2019) and VC datasets. FSRDC Project Number 2633 (CBDRB-FY23-P2633-R10186).

Figure 11: Average Employment and Median Payroll of Gazelles

by:

$$\tilde{g}_{st} := \left(\frac{X_{st}}{X_t}\right) g_{st}, \text{ where } X_t = \sum_s X_{st}.$$

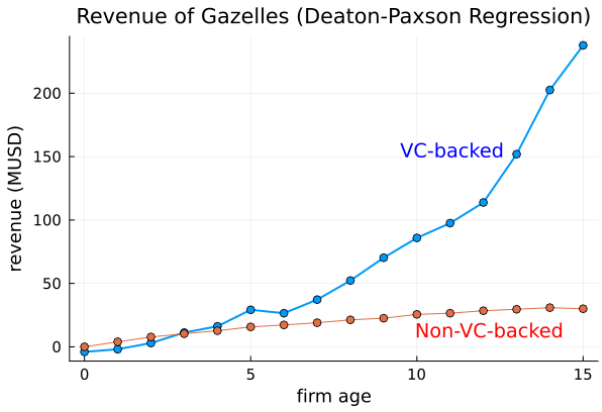
The term $\left(\frac{X_{st}}{X_t}\right)$ represents the employment or payroll share of group s in the economy.

Finally, the contribution of the group of firms over a long period of time is derived by taking the ratio between the geometric mean of the rates of contribution $1 + \tilde{g}_{st}$ and the geometric mean of aggregate growth rates $1 + g_t$. For example, if the long-run aggregate growth rate is 2% and the long-run contribution by group s is 0.2%, the group s accounts for 10% of the aggregate growth.

A.3 Additional Empirical Results

High Growth of VC-backed Gazelles This subsection shows the robustness of the high growth of VC-backed firms documented in Section 2.3. The left panel in Figure 11 compares the average employment of VC-backed and non-VC-backed employment Gazelles. The right panel in Figure 11 shows the growth trajectories of payroll Gazelles in terms of the (pseudo) median payroll. Figure 12 plots regression coefficients on age interacted with VC-backing status in a Deaton-Paxson regression, where firm revenue is the dependent variable, and year and NAICS (four digits) are control variables. This plot shows that the difference in growth trajectories between VC-backed and non-VC-backed firms is not solely explained by the difference in their industries. All figures show that VC-backed Gazelles attain high growth compared to non-VC-backed Gazelles.

Table 10 shows the regression results for the growth rate of firms in the sample of



*N*ote: In the Deaton-Paxson regression, the dependent variable is firm revenue. The independent variables are firm age interacted with VC-backed dummy, year, and NAICS four-digit codes. The coefficients on the age dummies are plotted in the figure. Data: LBD (1997-2019) and VC datasets. FSRDC Project Number 2633 (CBDRB-FY23-P2633-R10843).

Figure 12: Deaton-Paxson Regression for firm's revenue

Table 10: Growth rate of Gazelles (1980-2019)

	Dependent variable (%):	
	(1)	(2)
	emp growth	payroll growth
VC-backed	3.76 ***	3.02***
	(0.691)	(0.705)
firm age	–1.39***	–1.31***
	(0.0571)	(0.152)
log(emp)	–0.0603	
	(0.246)	
log(payroll)		0.251
		(0.246)
Year	Yes	Yes
Initial Year	Yes	Yes
NAICS (4 digits)	Yes	Yes
N	2,070,000	2,235,000
R^2	0.036	0.046

* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

*N*ote: The sample includes Gazelles established from 1980 to 2010 and aged from 0 to 20 in the years 1980–2019. The sample in column (1) includes employment Gazelles, while the sample in column (2) includes payroll Gazelles. Gazelles are firms with an average annual growth rate above 20% in the first five years and employment of 100 (or payroll of 100 times the average payroll per worker in the economy) or above by age 20. The dependent variables are the DHS growth rates of employment and payrolls, $g_{it} = \frac{E_{i,t} - E_{i,t-1}}{0.5*(E_{i,t}+E_{i,t-1})}$. Data: LBD (1980-2019) and VC datasets. FSRDC Project Number 2633 (CBDRB-FY23-P2633-R10843).

Table 11: Comparison of Exit Rate between VC-backed firms and Non-VC-backed firms

	Dependent variable: exit at $t+1$		
	(1)	(2)	(3)
	all firms	emp Gazelles	payroll Gazelles
VC-backed	–0.0248***	0.00104	0.0120***
	(0.000491)	(0.000643)	(0.000585)
Year	Yes	Yes	Yes
Age Dummy	Yes	Yes	Yes
NAICS (4 digits)	Yes	Yes	Yes
N	136,200,000	2,185,000	2,348,000
R^2	0.015	0.013	0.011

Standard errors in parentheses

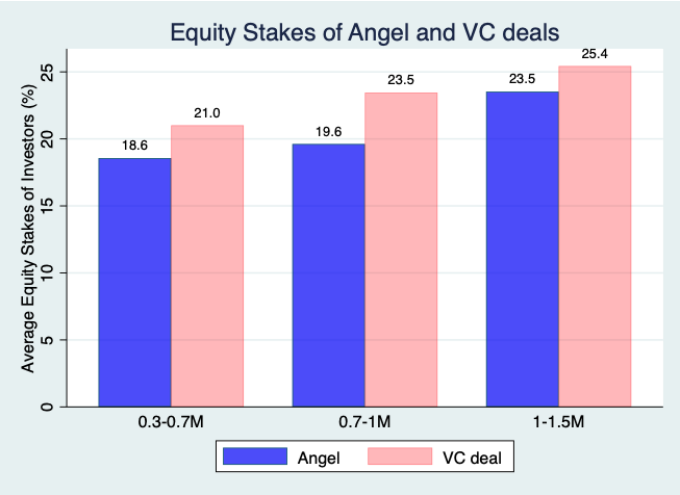
* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

*N*ote: The dependent variable (exit) is one if a firm exits in the next period. The sample includes firms established from 1980 to 2010 and aged from 0 to 20 in the years 1980–2019. Data: LBD (1980-2019) and VC datasets. FSRDC Project Number 2633 (CBDRB-FY23-P2633-R10843).

Gazelles. This regression addresses two issues in Figure 1. In the figure, the average firm size is computed among surviving firms. Also, the firm size may increase by acquiring another firm. Here, since the growth rate is measured in terms of a DHS growth rate, its value is –2 if the firm exits in the next period instead of being dropped from the sample. Additionally, the growth rate is robust to ownership changes since the firm growth is measured as the sum of the organic expansion of establishments that belong to the firm in the next period. After taking into account the two potential concerns (survival bias and ownership changes), VC-backed firms are shown to attain high growth compared to non-VC-backed firms. Note that after controlling for firm age, the coefficients on firm size measured by log(emp) and log(payroll) are statistically insignificant. This finding is consistent with the finding by Haltiwanger et al. (2013) that after controlling for firm age, firm growth is not systematically related to firm size (i.e., Gibrat's Law).

Finally, Table 11 compares the exit rate between VC-backed firms and non-VC-backed firms. Among firms aged 0 to 20, the exit rate of VC-backed firms is around 2.5% lower compared to non-VC-backed firms after controlling for year, age, and NAICS (four digits). In the sample of Gazelles, the difference in exit rates is statistically insignificant in the sample of employment Gazelles and is positive with around 1.2% in the sample of payroll Gazelles.

Equity Stakes in VC deals and Angel deals Figure 13 demonstrates the average equity stakes in VC and Angel financing. In the sample of first deals, VC investors acquire more equity stakes than Angel investors conditional on deal size categorized into three bins.



*N*ote: The first financing deals (either VC or Angel deals) for each company are included in the sample. Sample period: 1990-2021. Sample size: 4135. Data source: Pitchbook.

Figure 13: Equity Stake in VC deals and Angel deals

The difference ranges from 1.9% to 3.9%, depending on the size bin.

B Mathematical Derivations

B.1 Equity Stakes

This section derives the equivalence of two formulations, maximizing the firm's total value and maximizing the startup founder's value, along with the evolution of equity stakes owned by investors. This equivalence is an intuitive result, as investors and startup founders are both risk-neutral in the model.

The maximization problem of the firm value is examined first. I consider the case where the firm raises equity and distributes zero dividends ($d_t < 0$). The amount of money raised is denoted by $m_t = -d_t$. I rewrite the firm's maximization problem in equity financing in equation (9) while assuming no exogenous exit shocks to simplify the notation:

$$V_t^{\text{total}} = \max_{m_t} -(1 + c_{e,1})m_t + \frac{1}{1+r}\mathbb{E}\left[V_{t+1}^{\text{total}}\right] \tag{23}$$
$$\text{s.t. expend}_t = \text{rev}_t + m_t,$$

where $\text{expend}_t = Rk_{t+1} + w\ell_{t+1} + I_t + c_f + c_{e,0} + \mathbf{1}\{\text{VC}\}\bar{\pi}_{\text{vc}}$ and $\text{rev}_t = (\hat{\theta}_{n_t} + \epsilon_t)^{1-\eta}(k_t^{\alpha}\ell_t^{1-\alpha})^{\eta}$. The investors inject money m_t into the firm, which costs $(1 + c_{e,1})m_t$ due to the flotation costs $c_{e,1}m_t$. The discounted firm value after the investment, which is called post-money valuation, is given by $\frac{1}{1+r}\mathbb{E}\left[V_{t+1}^{\text{total}}\right]$.

Next, the startup founder's value is formulated as follows. The founder holds $(1 - s_t)$ fraction of the firm's pre-money valuation (i.e., firm value before raising equity financing) at time t, where $V_t^{\text{pre}} + X_t = V_t^{\text{post}}$ and $X_t = (1 + c_{e,1})m_t$. The pre-money valuation corresponds to V_t^{total} in the formulation above. After equity investment at time t, the founder holds $(1 - s_{t+1})$ fraction of the firm's post-money valuation. The founder's maximization problem is given by:

$$V_t^{\text{startup}} = \max_{m_t} (1 - s_{t+1}) \frac{1}{1+r} \mathbb{E}_t \left[V_{t+1}^{\text{pre}} \right] \tag{24}$$

$$\text{s.t. } s_{t+1} = s_t + (1 - s_t) \frac{X_t}{V_t^{\text{post}}} \tag{25}$$

$$\text{expend}_t = \text{rev}_t + m_t.$$

The founder owns $(1 - s_{t+1})$ fraction of the firm ownership after the financial investment at time t.

First, I verify the evolution of equity stakes in equation (25) using the investor's zero-profit condition. Old investors who acquired the equity stakes of the firm before time t hold s_t fraction of the firm's pre-money valuation at the beginning of the period. The value of their equity stakes should be the same after the firm raises additional equity financing in this period. By denoting their equity stakes after the equity investment in this period by $\tilde{s}_t$, the following equation holds:

$$s_t V_t^{\text{pre}} = \tilde{s}_t V_t^{\text{post}}.$$

New equity investors inject m_t into the firm at time t by paying $X_t = (1 + c_{e,1})m_t$ in exchange for the $\tilde{s}_{t+1}$ fraction of the firm's post-money valuation. Their zero-profit condition implies:

$$X_t = \tilde{s}_{t+1} V_t^{\text{post}}. \tag{26}$$

Hence, the total equity stakes of post-money valuation held by the old and new investors, denoted by s_{t+1}, are given by:

$$\begin{aligned} s_{t+1} &= \tilde{s}_t + \tilde{s}_{t+1} \\ &= s_t \frac{V_t^{\text{pre}}}{V_t^{\text{post}}} + \frac{X_t}{V_t^{\text{post}}} \\ &= s_t + (1 - s_t) \frac{X_t}{V_t^{\text{post}}}. \end{aligned}$$

Hence, equation (25) is obtained.

Second, I show the equivalence between the two maximization problems by substituting (25) into (24). Since the founder's equity stakes of the firm's post-money valuation, $1 - s_{t+1}$, are given by:

$$1 - s_{t+1} = (1 - s_t)\left(1 - \frac{X_t}{V_t^{\text{post}}}\right),$$

the founder's value at time t is given by:

$$V_t^{\text{startup}} = \max_{m_t}(1 - s_t)\left[1 - \frac{X_t}{V_t^{\text{post}}}\right]\frac{1}{1+r}\mathbb{E}\left[V_{t+1}^{\text{pre}}\right].$$

Since the firm distributes zero dividends in this period, the post-money valuation is:

$$V_t^{\text{post}} = \frac{1}{1+r}\mathbb{E}\left[V_{t+1}^{\text{pre}}\right].$$

Hence, the objective function (24) is written as:

$$V_t^{\text{startup}} = (1 - s_t)\left\{\max_{m_t} -(1 + c_{e,1})m_t + \frac{1}{1+r}\mathbb{E}\left[V_{t+1}^{\text{pre}}\right]\right\}$$
$$\text{s.t. expend}_t = \text{rev}_t + m_t.$$

It can be seen that maximizing the founder's value, which is $(1 - s_t)$ fraction of the firm's value, is equivalent to maximizing the firm's total value in (23).

B.2 Optimality Conditions

B.2.1 Economy without Financial Frictions

This subsection describes the optimality conditions with respect to capital inputs and innovation expenditures in an economy without financial frictions. Since the value function in this setup can be solved very quickly (i.e., less than one second to compute the stationary distribution given parameters), I calibrate this economy and obtain a reasonable initial guess of parameters for the economy with financial frictions described in Section 3.

Firms are endowed with growth potential $\hat{\theta}$ with corresponding productivity ladders $\{\hat{\theta}_n\}_{n=1,\cdots,N}$. The value function is given below. Since firms do not face financial frictions,

the cost of raising one unit of capital is one.

$$V(n_t, \epsilon_t, y_t, \text{fail}_t; \hat{\theta}) = y_t + \max\Bigg\{0, \max_{k_{t+1}, I_t} \; -\frac{R}{\alpha}k_{t+1} - I_t - c_f$$
$$+ \frac{1-\varphi}{1+r}\mathbb{E}_t\left[V_{t+1}(n_{t+1}, \epsilon_{t+1}, y_{t+1}, \text{fail}_{t+1}; \hat{\theta})\right] + \frac{\varphi}{1+r}\mathbb{E}_t\left[y_{t+1}\right]\Bigg\},$$
$$\text{where } y_{t+1} = (\hat{\theta}_{n_{t+1}} + \epsilon_{t+1})^{1-\eta}(k_{t+1}^{\alpha}\ell_{t+1}^{1-\alpha})^{\eta}$$
$$= (\hat{\theta}_{n_{t+1}} + \epsilon_{t+1})^{1-\eta}\left(\frac{1-\alpha}{\alpha}\frac{R}{w}\right)^{(1-\alpha)\eta} k_{t+1}^{\eta}. \tag{27}$$

In equation (27), labor input is substituted, using the FOC with respect to (ℓ_{t+1}, k_{t+1}):

$$\ell_{t+1} = \frac{1-\alpha}{w}\frac{R}{\alpha}k_{t+1}. \tag{28}$$

The FOC of the objective function with respect to k_{t+1} is:

$$\underbrace{\frac{R}{\alpha}}_{\text{MC}} = \frac{1-\varphi}{1+r}\mathbb{E}_t\left[V_{k_{t+1}}(n_{t+1}, \epsilon_{t+1}, y_{t+1}, \text{fail}_{t+1}; \hat{\theta})\right] + \frac{\varphi}{1+r}E_t\left[\frac{\partial}{\partial k_{t+1}}y_{t+1}\right],$$
$$\text{where } V_k(n_t, \epsilon_t, y_t, \text{fail}_t; \hat{\theta}) = \frac{\partial}{\partial k_t}y_t = (\hat{\theta}_{n_t} + \epsilon_t)^{1-\eta}\left(\frac{1-\alpha}{\alpha}\frac{R}{w}\right)^{(1-\alpha)\eta}\eta k_t^{\eta-1}.$$

Hence, the optimal k_{t+1} is derived as:

$$k_{t+1} = \left\{\frac{\alpha}{R}\left(\frac{1-\alpha}{\alpha}\frac{R}{w}\right)^{(1-\alpha)\eta}\frac{\eta}{1+r}\mathbb{E}_{t+1}\left[(\theta_{n_{t+1}} + \epsilon_{t+1})^{1-\eta}\right]\right\}^{\frac{1}{1-\eta}}. \tag{29}$$

The FOC with respect to I_t is:

$$1 = \frac{1-\varphi}{1+r}\frac{\partial \varsigma_{t+1}}{\partial I_t}\mathbb{E}_t\left[V_{t+1}(n_t+1, \epsilon_{t+1}, y_{t+1}, \text{fail}_{t+1}=0) - V_{t+1}(n_t, \epsilon_{t+1}, y_{t+1}, \text{fail}_{t+1}=1)\right]$$
$$+\frac{\varphi}{1+r}\frac{\partial \varsigma_{t+1}}{\partial I_t}\mathbb{E}_t\left[y_{t+1}(n_t+1, \epsilon_{t+1}, y_{t+1}) - y_{t+1}(n_t, \epsilon_{t+1}, y_{t+1})\right]. \tag{30}$$

Since the derivative of innovation success rate (ς_{t+1}) with respect to innovation expenditures (I_t) is:

$$\frac{\partial \varsigma_{t+1}}{\partial I_t} = \varsigma_{t+1}(1-\varsigma_{t+1})\frac{\gamma_1}{I_t}, \tag{31}$$

the optimal I_t follows:

$$1 = \frac{\gamma_1 \varsigma^*(1-\varsigma^*)}{(1+r)I_t^*} \Bigg\{ (1-\varphi)\mathbb{E}_t \left[V(n_t+1, \epsilon_{t+1}, y_{t+1}, \text{fail}_{t+1}=0) - V(n_t, \epsilon_{t+1}, y_{t+1}, \text{fail}_{t+1}=1) \right] + \varphi(\hat{\theta}_{n_t+1} - \hat{\theta}_{n_t}) \left(\frac{1-\alpha}{\alpha} \frac{R}{w} \right)^{(1-\alpha)\eta} (k_{t+1})^{\eta} \Bigg\}. \tag{32}$$

The optimal (k_{t+1}, I_t) is a solution to a system of two equations (29) and (32) and can be computed with a non-linear solver.

B.2.2 Equity Financing

The value function of equity-backed firms is formulated in Section 3.2. The Lagrangian formulation of the continuation value of equity-financed firms ($j \in \{\text{VC}, \text{Angel}\}$) is as follows:

$$\begin{aligned} V_t^{j,C}(n_t, \epsilon_t, o_t, \text{fail}_t; \hat{\theta}) = & \max_{d_t, k_{t+1}, I_t} d_t + \mathbf{1}\{d_t < 0\}(d_t c_{e,1}) \\ & + \frac{1-\varphi}{1+r}\mathbb{E}_t \left[V_{t+1}^j(n_{t+1}, \epsilon_{t+1}, o_{t+1}, \text{fail}_{t+1}; \hat{\theta}) \right] + \frac{\varphi}{1+r}\mathbb{E}_t \left[o_{t+1} \right] \\ & + \mu_t \left[o_t - \frac{R}{\alpha} k_{t+1} - I_t - c_f - c_{e,0} - \mathbf{1}\{\text{j=VC}\}\bar{\pi}_{\text{vc}} - d_t \right] \end{aligned} \tag{33}$$

$$\text{s.t. } o_t = (\hat{\theta}_{n_t} + \epsilon_t)^{1-\eta} \left(\frac{1-\alpha}{\alpha} \frac{R}{w} \right)^{(1-\alpha)\eta} k_t^{\eta}. \tag{34}$$

Note that the labor input (ℓ_{t+1}) is substituted by equation (28). The FOCs with respect to the three control variables are:

$$[d_t]: \quad (1 + \mathbf{1}\{d_t < 0\}c_{e,1} - \mu_t)\, d_t = 0 \tag{35}$$

$$[k_{t+1}]: \quad \frac{1-\varphi}{1+r}\mathbb{E}_t \left[\frac{\partial V_{t+1}}{\partial k_{t+1}} \right] + \frac{\varphi}{1+r}\mathbb{E}_t \left[\frac{\partial y_{t+1}}{\partial k_{t+1}} \right] = \frac{R}{\alpha}\mu_t \tag{36}$$

$$[I_t]: \quad \frac{\partial \varsigma_{t+1}}{\partial I_t} \left\{ \frac{1-\varphi}{1+r}\, \mathbb{E}_t \left[V(n+1) - V(n) \right] + \frac{\varphi}{1+r}\mathbb{E}_t \left[o_{t+1}(n+1) - o_{t+1}(n) \right] \right\} = \mu_t. \tag{37}$$

The marginal cost of raising additional financial capital is given by μ_t, instead of 1, in equity financing. If $d_t > 0$, since the firm is not liquidity constrained, $\mu_t = 1$. If $d_t < 0$, since the firm relies on equity financing, $\mu_t = 1 + c_{e,1}$. If $d_t = 0$, the Lagrangian multiplier μ_t satisfies $\mu_t \geq 1$, (36), and (37), while (k_{t+1}, I_t) follows the budget constraint. The

derivative of the value function is given by:

$$\frac{\partial V_t}{\partial k_t} = \mu_t \frac{\partial y_t}{\partial k_t}. \tag{38}$$

By using:

$$\frac{\partial y_t}{\partial k_t} = (\hat{\theta}_{n_t} + \epsilon_t)^{1-\eta} \left(\frac{1-\alpha}{\alpha} \frac{R}{w} \right)^{(1-\alpha)\eta} \eta k_t^{\eta-1}$$

and equation (31), the conditions are summarized as follows:

$$[d_t]: \quad \mu_t \begin{cases} = 1 & \text{if } d_t > 0 \\ \geq 1 & \text{if } d_t = 0 \\ = 1 + c_{e,1} & \text{if } d_t < 0 \end{cases} \tag{39}$$

$$\begin{aligned} [k_{t+1}]: \quad & \left\{ \frac{1-\varphi}{1+r} \mathbb{E}_t \left[(\hat{\theta}_{n_{t+1}} + \epsilon_{t+1})^{1-\eta} \mu_{t+1} \right] + \frac{\varphi}{1+r} \mathbb{E}_t \left[(\hat{\theta}_{n_{t+1}} + \epsilon_{t+1})^{1-\eta} \right] \right\} \\ & \times \left(\frac{1-\alpha}{\alpha} \frac{R}{w} \right)^{(1-\alpha)\eta} \eta k_t^{\eta-1} = \frac{R}{\alpha} \mu_t \end{aligned} \tag{40}$$

$$\begin{aligned} [I_t]: \quad & \frac{\gamma_1 \varsigma_{t+1}^* (1 - \varsigma_{t+1}^*)}{I_t} \Big\{ \frac{1-\varphi}{1+r} \mathbb{E}_t \left[V(n_t + 1) - V(n_t) \right] \\ & + \frac{\varphi}{1+r} \mathbb{E}_t \left[o_{t+1}(n_t + 1) - o_{t+1}(n_t) \right] \Big\} = \mu_t. \end{aligned} \tag{41}$$

To numerically solve this problem, the optimal choice of (k_{t+1}, I_t, d_t) is computed for each case $(d_t > 0, d_t = 0, d_t < 0)$. The choice satisfying (39)–(41) and the budget constraint and maximizing the continuation value is chosen as a solution. The value function is iterated until it converges.

B.2.3 Bank Financing

The value function in bank financing is formulated in Section 3.3. The continuation value is given by equation (14) with the price of debt given by equation (13). Note that the FOCs with respect to (ℓ_{t+1}, k_{t+1}) results in equation (28). Thus, one control variable (ℓ_{t+1}) is eliminated from the value function iteration. Moreover, by substituting d_t and $q_t(\cdot)$ using the budget constraint and the price of debt (13), the continuation value can

be expressed without the price of debt (q_t) as follows:

$$
\begin{aligned}
V_t^{\text{Bank,C}}(n_t, \epsilon_t, o_t, \text{fail}_t; \hat{\theta}) = & \max_{k_{t+1}, I_t, b_{t+1}} o_t - \frac{R}{\alpha} k_{t+1} - I_t - c_f \\
& + \frac{1}{1+r} \mathbb{E}_t \Bigg[(1-\varphi) \Big\{ \mathbf{1}\left\{ V_{t+1}^{\text{bank}}(\cdot) \geq 0 \right\} \left(V_{t+1}^{\text{bank}}(n_{t+1}, \epsilon_{t+1}, o_{t+1}, \text{fail}_{t+1}) + b_{t+1} \right) \\
& \quad + \mathbf{1}\left\{ V_{t+1}^{\text{bank}}(\cdot) < 0 \wedge o_{t+1} \geq 0 \right\} y_{t+1} + \left\{ V_{t+1}^{\text{bank}}(\cdot) < 0 \wedge o_{t+1} < 0 \right\} \left(y_{t+1} - c_d b_{t+1} \right) \Big\} \\
& + \varphi \Big\{ \mathbf{1}\left\{ o_{t+1} \geq 0 \right\} y_{t+1} + \mathbf{1}\left\{ o_{t+1} < 0 \right\} \left(y_{t+1} - c_d b_{t+1} \right) \Big\} \Bigg], \qquad (42) \\
& \text{where } y_{t+1} = (\hat{\theta}_{n_{t+1}} + \epsilon_{t+1})^{1-\eta} \left(\frac{1-\alpha}{\alpha} \frac{R}{w} \right)^{(1-\alpha)\eta} k_{t+1}^{\eta}.
\end{aligned}
$$

Therefore, the continuation value $V_t^{\text{Bank,C}}(\cdot)$ can be computed without separately computing the pricing function $q_t(\cdot)$.[60]

Lagrangian Formulation The value function is computed based on the Lagrangian formulation:

$$
\begin{aligned}
V_t^{\text{Bank},C}(o_t, \epsilon_t, \text{fail}_t, n_t; \hat{\theta}) = & \max_{d_t \geq 0, k_{t+1}, I_t, b_{t+1}} d_t + \frac{1}{1+r} \mathbb{E}\left[V_{t+1}^{\text{bank}}(o_{t+1}, \epsilon_{t+1}, \text{fail}_{t+1}, n_{t+1}; \hat{\theta}) \right] \\
& + \mu_t \left[o_t + q_t(b_{t+1}, k_{t+1}, I_t) b_{t+1} - d_t - \frac{R}{\alpha} k_{t+1} - I_t - c_f \right].
\end{aligned}
$$

[60] The derivation in a case without an exogenous shock is as follows. This is done by substituting dividend (d_t) using a budget constraint.

$$
\begin{aligned}
V_t^{\text{bank}}(n_t, \epsilon_t, o_t, \text{fail}_t; \hat{\theta}) = & \max_{k_{t+1}, I_t, b_{t+1}} o_t \\
& + \frac{1}{1+r} \mathbb{E}_t \left[\mathbf{1}\left\{ \Delta_{t+1} = 0 \right\} + \mathbf{1}\left\{ \Delta_{t+1} = 1 \right\} \left(\frac{y_{t+1}}{b_{t+1}} - c_d \right) \right] b_{t+1} - \frac{R}{\alpha} k_{t+1} - I_t - c_f \\
& + \frac{1}{1+r} \mathbb{E}_t \left[\mathbf{1}\left\{ \text{continue}_{t+1} \right\} V_t^{\text{bank}}(n_{t+1}, \epsilon_{t+1}, o_{t+1}, \text{fail}_{t+1}) + \mathbf{1}\left\{ \text{exit}_{t+1} \right\} o_{t+1} + \mathbf{1}\left\{ \Delta_{t+1} = 1 \right\} 0 \right] \\
= & \max_{k_{t+1}, I_t, b_{t+1}} o_t - \frac{R}{\alpha} k_{t+1} - I_t - c_f \\
& + \frac{1}{1+r} \mathbb{E}_t \Big[\mathbf{1}\left\{ \text{cont}_{t+1}(n_{t+1}, \epsilon_{t+1}, o_{t+1}, \text{fail}_{t+1}) \right\} \left(V_{t+1}^{\text{bank}}(n_{t+1}, \epsilon_{t+1}, o_{t+1}, \text{fail}_{t+1}) + b_{t+1} \right) \\
& + \mathbf{1}\left\{ \text{exit}_{t+1}(n_{t+1}, \epsilon_{t+1}, o_{t+1}, \text{fail}_{t+1}) \right\} y_{t+1} + \mathbf{1}\left\{ \Delta_{t+1}(n_{t+1}, \epsilon_{t+1}, o_{t+1}, \text{fail}_{t+1}) = 1 \right\} \left(y_{t+1} - c_d b_{t+1} \right) \Big].
\end{aligned}
$$

FOCs are given by:

$$[d_t]: \quad (1-\mu_t)\, d_t = 0 \tag{43}$$

$$[k_{t+1}]: \quad \mu_t \left[\frac{R}{\alpha} - \frac{\partial q_t(\cdot)}{\partial k_{t+1}}\, b_{t+1}\right] = \frac{1}{1+r}\mathbb{E}_t\left[\frac{\partial}{\partial k_{t+1}} V_{t+1}^{\text{bank}}(\cdot)\right] \tag{44}$$

$$[I_t]: \quad \mu_t \left[1 - \frac{\partial q_t(\cdot)}{\partial I_t}\, b_{t+1}\right] = \frac{1}{1+r}\mathbb{E}_t\left[\frac{\partial}{\partial I_t} V_{t+1}^{\text{bank}}(\cdot)\right] \tag{45}$$

$$[b_{t+1}]: \quad \mu_t \left[q_t(\cdot) + \frac{\partial q_t(\cdot)}{\partial b_{t+1}}\, b_{t+1}\right] = \frac{-1}{1+r}\mathbb{E}_t\left[\frac{\partial}{\partial b_{t+1}} V_{t+1}^{\text{bank}}(\cdot)\right] \tag{46}$$

By defining $\mu_{t+1}(o_{t+1}, \epsilon_{t+1}, \text{fail}_{t+1}, n_{t+1}; \hat{\theta}) = 0$ if $\Delta_{t+1}(o_{t+1}, \epsilon_{t+1}, \text{fail}_{t+1}, n_{t+1}; \hat{\theta}) = 1$ (i.e., Lagrangian multiplier is zero in a state the firm defaults), I obtain:

$$\frac{\partial}{\partial k_{t+1}} V_{t+1}^{\text{bank}}(\cdot) = \mu_{t+1}\frac{\partial o_{t+1}}{\partial k_{t+1}}$$
$$\frac{\partial}{\partial b_{t+1}} V_{t+1}^{\text{bank}}(\cdot) = -\mu_{t+1}$$

A useful Lemma is shown below:

Lemma 1.
(a) If $P(\Delta_{t+1}) = 0$*,* $\mu_t = \mathbb{E}_t[\mu_{t+1}] \geq 1$.
(b) If $P(\Delta_{t+1}) > 0$*,* $\mu_t > 1$.

Proof. (a) If $P(\Delta_{t+1}) = 0$ (zero probability of default in the next period), the price of debt is given by (due to the zero profit condition):

$$q_t(\cdot) = \frac{1}{1+r}.$$

Then, (46) implies

$$\mu_t = \mathbb{E}_t[\mu_{t+1}].$$

When a firm does not default ($\Delta_{t+1} = 0$), the Lagrangian multiplier is greater than 1 ($\mu_{t+1} \geq 1$) due to the Kuhn-Tucker condition (43).

(b) Equation (46) implies that

$$\mu_t = \frac{\frac{1}{1+r}\mathbb{E}_t\left[\mu_{t+1}\right]}{q_t(\cdot) + \frac{\partial q_t(\cdot)}{\partial b_{t+1}}\, b_{t+1}}. \tag{47}$$

By rewriting $q_t(\cdot)$ as:

$$q_t(b_{t+1}, k_{t+1}, I_t) = \frac{1}{1+r}\mathbb{E}_t\left[1 - \mathbf{1}\left\{\Delta_{t+1}=1\right\}\left(1 - \frac{y_{t+1}}{b_{t+1}} + c_d\right)\right], \tag{48}$$

I obtain:

$$\begin{aligned}\frac{\partial q_t(\cdot)}{\partial b_{t+1}} =& \frac{-1}{1+r}\mathbb{E}_t\left[\mathbf{1}\left\{\Delta_{t+1}=1\right\}\frac{y_{t+1}}{(b_{t+1})^2}\right] \\ &- \frac{1}{1+r}\mathbb{E}_t\left[\mathbf{1}\left\{\Delta_{t+1}=1\right\}\left(1 - \frac{y_{t+1}}{b_{t+1}} + c_d\right)\right]\frac{\partial}{\partial b_{t+1}}P(\Delta_{t+1}=1).\end{aligned}$$

The second term is positive, which will be denoted by $\frac{1}{(1+r)b_{t+1}}\tilde{\Delta} \geq 0$. Equation (47) is given by:

$$\begin{aligned}\mu_t &= \frac{\frac{1}{1+r}\mathbb{E}_t\left[\mu_{t+1}\right]}{\frac{1}{1+r}\mathbb{E}_t\left[1 - \mathbf{1}\left\{\Delta_{t+1}=1\right\}\left(1 - \frac{y_{t+1}}{b_{t+1}} + c_d\right)\right] - \frac{1}{1+r}\mathbb{E}_t\left[\mathbf{1}\left\{\Delta_{t+1}=1\right\}\frac{y_{t+1}}{b_{t+1}}\right] - \frac{1}{1+r}\tilde{\Delta}} \\ &= \frac{\mathbb{E}_t\left[\mu_{t+1}\right]}{1 - (1+c_d)P(\Delta_{t+1}) - \tilde{\Delta}}.\end{aligned} \tag{49}$$

Because $\mathbb{E}_t\left[\mu_{t+1}\right]$ is larger than the sum of two cases, $1 \cdot P(\Delta_{t+1}=0) + 0 \cdot P(\Delta_{t+1}=1)$,

$$\mu_t \geq \frac{1 - P(\Delta_{t+1}=1)}{1 - (1+c_d)P(\Delta_{t+1}) - \tilde{\Delta}} > 1.$$

□

From Property (b), if $P(\Delta_{t+1}=1) > 0$ (or $q_t(\cdot) < \frac{1}{1+r}$), $d_t = 0$ by the Kuhn-Tucker condition (43). Also, $\mu_t = 1$ implies $P(\Delta_{t+1}=1) = 0$ and $q_t(\cdot) = \frac{1}{1+r}$.[61]

Value Function Iteration (VFI) The value function is computed separately at the last step of the productivity ladder ($n = N$), where firms do not invest in innovation, and at $n < N$, where firms invest in innovation. In addition, the firm's decision is separately analyzed for two cases: $d_t > 0$ and $d_t = 0$.

If $d_t > 0$ (i.e., firms distribute dividends), the Lagrangian multiplier is one ($\mu_t = 1$) implied by the Kuhn-Tucker condition (43). If $d_t = 0$, the Lagrangian multiplier (the

[61] If $P(\Delta_{t+1}=1) = 0$ and $q_t(\cdot) = \frac{1}{1+r}$, firms could borrow money from banks at a risk-free rate and distribute it as dividend, which does not change the firm value. Such solutions are eliminated by assumption in the value function iteration. On the other hand, if firms may face a bank loan rate higher than a risk-free rate in the next period, firms have an incentive to save this period. Firms would not choose $d_t > 0$ if $\mathbb{E}\left[\mu_{t+1}\right] > 1$, since firms increase savings ($b_t < 0$) until $d_t = 0$ or $\mathbb{E}[\mu_{t+1}] = 1$ is satisfied. In the firm dynamics literature, the issue of indeterminacy of debt is addressed, for instance, by introducing tax incentives for debt (Corbae and D'Erasmo, 2021), impatient households (Cooley and Quadrini, 2001), or positive default probability on all occasions (Khan et al., 2014).

shadow price of cash flow) is larger than or equal to one. Note that the pricing function $q_t(\cdot)$ is computed given the value function in the next period:

$$\begin{aligned} q_t(\cdot) &= \frac{1}{1+r}\left[P(\Delta_{t+1}=0) + P(\Delta_{t+1}=1)\frac{\mathbb{E}_t\left[y_{t+1}-c_d b_{t+1}|\Delta_{t+1}=1\right]}{b_{t+1}}\right] \\ &= \frac{1}{1+r}\mathbb{E}_t\Big[(1-\varphi)\big(\mathbf{1}\left\{V_{t+1}(\cdot)\geq 0\right\} + \mathbf{1}\left\{V_{t+1}(\cdot)<0 \wedge o_{t+1}\geq 0\right\} \\ &\qquad + \mathbf{1}\left\{V_{t+1}(\cdot)<0 \wedge o_{t+1}<0\right\}\left(\frac{y_{t+1}}{b_{t+1}}-c_d\right)\big) \\ &\qquad + \varphi\left(\mathbf{1}\left\{o_{t+1}\geq 0\right\} + \mathbf{1}\left\{o_{t+1}<0\right\}\left(\frac{y_{t+1}}{b_{t+1}}-c_d\right)\right)\Big]. \end{aligned} \tag{50}$$

The budget constraint is given by:

$$d_t + \frac{R}{\alpha}k_{t+1} + I_t + c_f = o_t + q_t(b_{t+1}, k_{t+1}, I_t)b_{t+1}. \tag{51}$$

I. Consider the case at $n = N$. In this case, since firms are in the last step of the productivity ladder, they do not invest in innovation, and a set of state variables is given by (o_t, ϵ_t). In each iteration, the optimal choice of (k_t, d_t, b_{t+1}) and the value function $V_t(\cdot)$ are computed, given the value function in the next period $V_{t+1}(\cdot)$, Lagrangian multiplier $\mu_t(\cdot)$, and pricing function $q_t(\cdot)$.

I–(i). A case with $d_t > 0$ and an unconstrained solution k_t^*:

If a firm has large enough cash flow o_t with Lagrangian multiplier $\mu_t = 1$, the unconstrained solution of capital k_{t+1}^* is chosen:

$$k_{t+1}^* = \mathbb{E}_t\left[(\hat{\theta}_N + \epsilon_{t+1})^{1-\eta}\right]^{\frac{1}{1-\eta}}\left\{\frac{\eta}{1+r}\left(\frac{\alpha}{R}\right)^{1-(1-\alpha)\eta}\left(\frac{1-\alpha}{w}\right)^{(1-\alpha)\eta}\right\}^{\frac{1}{1-\eta}}. \tag{52}$$

For each value of cash flow (o_t), the unconstrained solution of capital satisfies the budget constraint if:

$$o_t > \left(\frac{R}{\alpha}\right)k_{t+1}^* + c_f. \tag{53}$$

If this is the case, the solution is given by:

$$\begin{aligned} V_t(\cdot) &= d_t + (1-\varphi)\frac{1}{1+r}\mathbb{E}\left[V_{t+1}(o_{t+1}, \epsilon_{t+1}, \text{fail}_{t+1}=0, n_{t+1}=N)\right] + \varphi\frac{1}{1+r}\mathbb{E}\left[o_{t+1}\right], \\ d_t &= o_t - \left(\frac{R}{\alpha}\right)k_{t+1}^* - c_f, \\ b_{t+1} &= 0, \text{ and } k_{t+1} = k_{t+1}^*. \end{aligned}$$

I–(ii). A case with $d_t = 0$ and $b_t > 0$:
If the inequality (53) does not hold, a firm cannot purchase sufficient capital without raising a bank loan. The optimal amount of capital is chosen subject to the budget constraint (51). The dept pricing function $q_t(\cdot)$ is determined by equation (50), and the size of the debt and capital expenditures are determined to satisfy the budget constraint. The firm value is computed with several values of k_{t+1} from grid points, and the optimal k_{t+1} (and corresponding b_{t+1} determined by the budget constraint) that maximizes the firm value is chosen. Given the optimal (k_{t+1}, b_{t+1}) and $d_t = 0$, the firm value is given by:

$$V_t(\cdot) = (1-\varphi)\frac{1}{1+r}\mathbb{E}\left[V_{t+1}(o_{t+1}, \epsilon_{t+1}, \text{fail}_{t+1} = 0, n_{t+1} = N)\right] + \varphi\frac{1}{1+r}\mathbb{E}\left[o_{t+1}\right]$$

with $o_{t+1} = y_{t+1} - b_{t+1}$.

If the firm value is negative with any choice of k_{t+1} and cash flow is negative ($o_t < 0$), the firm defaults with the outstanding bank loan and exits the market. If cash flow is positive, $o_t > 0$, but the firm value is less than the cash flow with any choice capital, the firm exits the market after fully repaying the loan.

The value function is iterated, meaning that the value function $V_t(\cdot)$ is computed given an updated value function in the next period $V_{t+1}(\cdot)$, until it converges.[62]
II. Consider the case at $n < N$. In this case, firms invest in innovation.
II-(i). A case with $d_t > 0$ and an unconstrained solution (k_t^*, I_t^*):
The unconstrained solution (k_t^*, I_t^*) follows a system of two equations. The first equation is the same as equation (52), where the realization of productivity $z_{t+1} = \hat{\theta}_{n_{t+1}} + \epsilon_{t+1}$ involves uncertainty on innovation outcome and an exogenous productivity shock:

$$\mathbb{E}_t\left[z_{t+1}^{1-\eta}\right] = \mathbb{E}_t\left[\sigma_{t+1}(\theta_{n_t+1} + \epsilon_{t+1})^{1-\eta} + (1-\sigma_{t+1})(\theta_{n_t} + \epsilon_{t+1})^{1-\eta}\right]. \tag{54}$$

The FOC with respect to innovation investment (I_t) is given in equation (45):

$$\mu_t\left[1 - \frac{\partial q_t(\cdot)}{\partial I_t}\, b_{t+1}\right] = \frac{1}{1+r}\mathbb{E}_t\left[\frac{\partial}{\partial I_t}V_{t+1}^{\text{bank}}(\cdot)\right]$$

Using derivative of the innovation function equation (31), the two terms inside the equa-

[62] The Lagrangian multiplier and pricing function are updated once in several iterations so that the value function iteration is stable. The value function is iterated several times with the same choice of control variables before reoptimizing the control variables to achieve faster convergence.

tion (45) are derived as:[63]

$$\frac{\partial q_t(\cdot)}{\partial I_t} = \frac{-\gamma_1 \varsigma^*_{t+1}(1-\varsigma^*_{t+1})}{(1+r)I_t}\mathbb{E}_t\left[\Delta_{t+1(n_t+1)}\left(1-\frac{y_{t+1}(n_t+1)}{b_{t+1}}+c_d\right)-\Delta_{t+1(n_t)}\left(1-\frac{y_{t+1}(n_t)}{b_{t+1}}+c_d\right)\right]$$

$$\mathbb{E}_t\left[\frac{\partial}{\partial I_t}V^{\text{bank}}_{t+1}(\cdot)\right] = \varsigma^*_{t+1}(1-\varsigma^*_{t+1})\frac{\gamma_1}{I_t}\mathbb{E}_t\left[V^{\text{bank}}_{t+1}(\text{fail}_{t+1}=0,n_t+1)-V^{\text{bank}}_{t+1}(\text{fail}_{t+1}=1,n_t)\right]$$

The unconstrained solution (k^*_t, I^*_t) is obtained by solving a system of equations (52) and (45) with $\mu_t = 1$. If it satisfies the budget constraint given the firm's cash flow:

$$o_t > \left(\frac{R}{\alpha}\right)k^*_t + I^*_t + c_f, \tag{55}$$

the solution is given by:

$$\begin{aligned} V_t(\cdot) &= d_t + (1-\varphi)\frac{1}{1+r}\mathbb{E}\left[V_{t+1}(o_{t+1},\epsilon_{t+1},\text{fail}_{t+1},n_{t+1})\right] + \varphi\frac{1}{1+r}\mathbb{E}\left[o_{t+1}\right], \\ d_t &= o_t - \left(\frac{R}{\alpha}\right)k^*_{t+1} - I^*_t - c_f, \\ k_{t+1} &= k^*_{t+1}, I_t = I^*_t, \text{ and } b_{t+1} = 0. \end{aligned}$$

II-(ii). A case with $d_t = 0$ and $b_t > 0$:
If the unconstrained solution is not feasible with the given cash flow o_t, firms raise a bank loan and optimally choose capital and innovation expenditure subject to the budget constraint (51). The firm value is evaluated at several values of k_t from grid points, and the optimal k_t that maximizes the firm value is chosen. For each k_t, the optimal (I_t, b_{t+1}) is solved to satisfy the FOC (45) and the budget constraint (51). The system of two equations is solved with NLsolve and Optim commands in Julia.[64] Given the optimal choice of (k_t, I_t, b_{t+1}), the value function is updated and iterated until it converges. The value function is iterated several times with the same choice of control variables before reoptimizing the control variables. Note that solving the value function iteration at all steps of the productivity ladder $\{\hat{\theta}_n\}_{n=1,\cdots,N}$ takes a few minutes.

B.3 Additional Discussions

B.3.1 Additional Quantitative Results

Figure 14 shows the relation between a firm's growth potential (θ) and its bank default rate at age 2 (one period after its birth). Although high-growth and medium-potential

[63] $q_t(\cdot)$ is given by (48): $q_t(b_{t+1}, k_{t+1}, I_t) = \frac{1}{1+r}\mathbb{E}_t\left[1-\mathbf{1}\left\{\Delta_{t+1}=1\right\}\left(1-\frac{y_{t+1}}{b_{t+1}}+c_d\right)\right]$

[64] Several efforts are made to speed up computation: (i) the solution from the last iteration is used as an

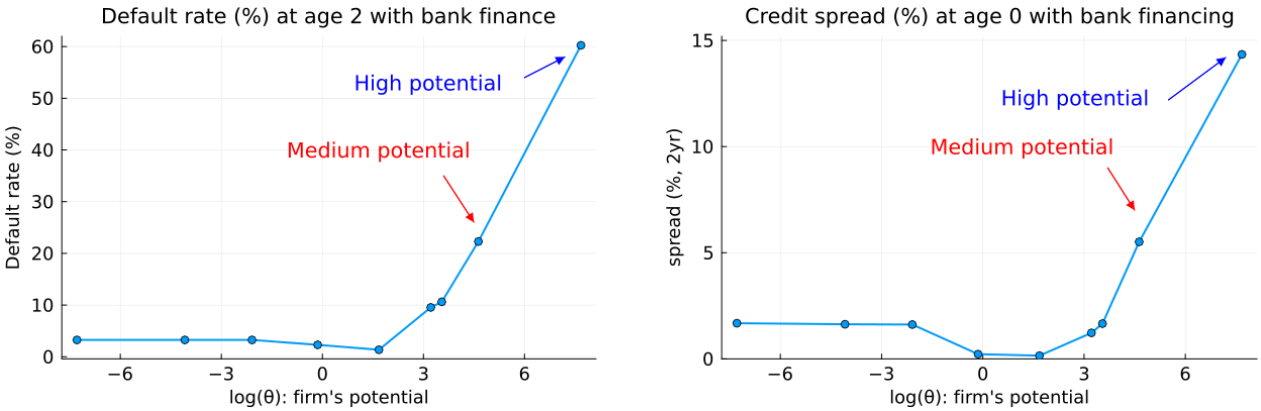


Figure 14: Default rate and Credit Spread for each θ

firms optimally choose VC financing and Angel financing in equilibrium, respectively, the figure shows the hypothetical bankruptcy rate if all firms choose bank financing. High-potential and medium-potential firms will default at a higher rate than lower-potential firms because higher-potential firms borrow more to finance innovation, but the innovation outcome is uncertain. Because of the high default rate, higher-potential firms face a higher bank loan rate, as seen in the higher credit spread (i.e., bank default rate minus risk-free rate) in the right panel.

Figure 15 shows the lifecycle dynamics of innovation success rate and the measure of survivors for each type of firm. The calibrated model implies that higher-potential firms have a lower innovation success rate than lower-potential firms. In the right panel, around half of VC-backed firms exit at age 2. This is because the value of VC-backed firms that failed in innovation is below zero, given the management fees paid to the venture capitalists. Among the bank-backed firms, higher-potential firms are more likely to survive, given their higher firm values. The panel at the bottom shows that high-potential firms invest more intensively in innovation, measured by innovation expenditures divided by firm value.

B.3.2 Note on the Tauchen Method

An exogenous productivity shock $\log(\epsilon_t)$ follows an $\mathrm{AR}(1)$ process. In the VFI, the stochastic process is approximated by the Tauchen's method. To reduce the computation time, a small number of grid points for $\log(\epsilon_t)$ is chosen (e.g. $N_\epsilon = 5$). In the standard Tauchen's method with a small number of grid points, the probability distribution across grid points is highly unequal, and the values of ϵ may not change frequently across periods. This

initial value, (ii) a value of k_{t+1} in the grid is eliminated from the potential solution if it cannot satisfy the budget constraint with the highest value of $q(\cdot)b_{t+1}$, (iii) once the value function passes the modal point with respect to k_{t+1} (firm value first increases with k_{t+1} and declines), a firm value at higher k_{t+1} is not evaluated.

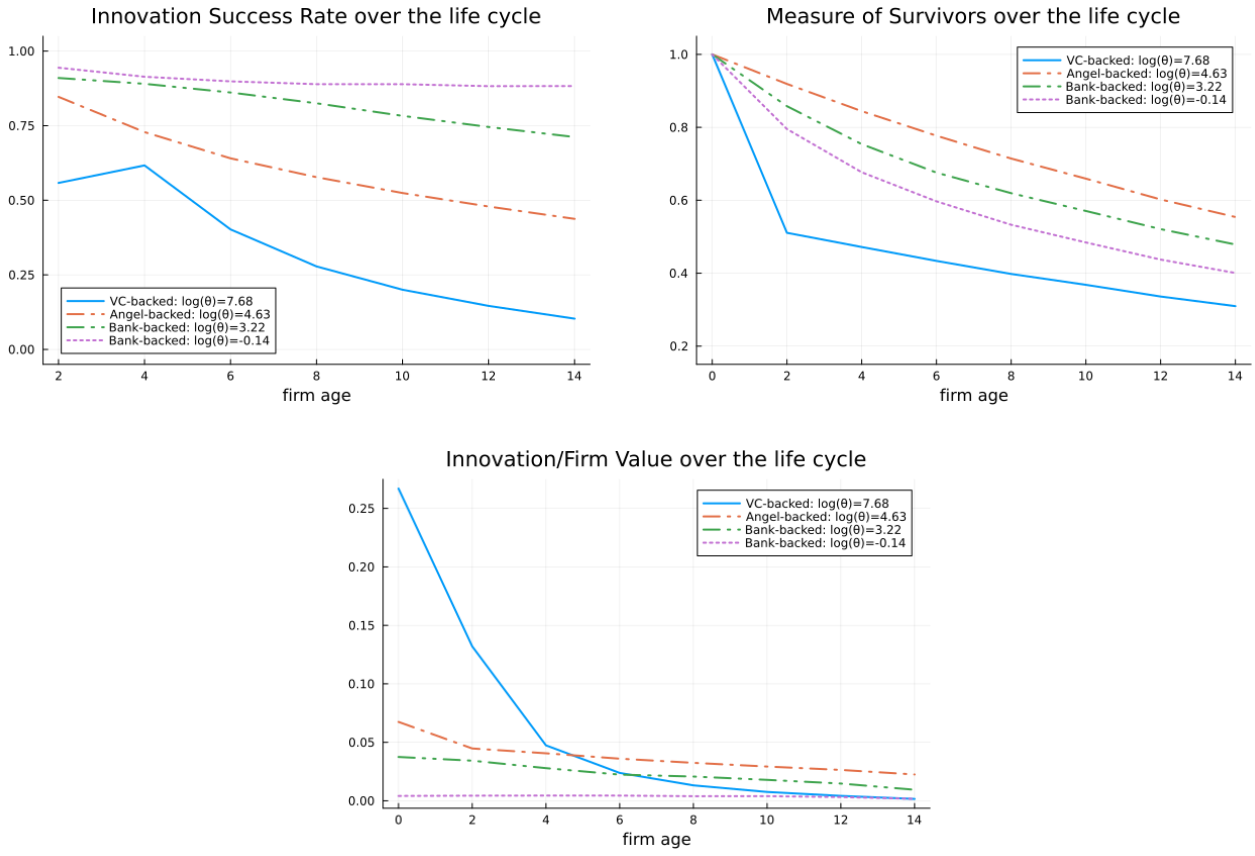


Figure 15: Lifecycle Dynamics

could lead to an unreasonably low exit rate in the model. To address this issue, the grid points are chosen so that the probability of each grid point in the stationary distribution is approximately equal.

To be specific, the grid points of $\log(\bar{\epsilon}_i)$ for $i = 1, \cdots, N_\epsilon$ are set such that the following relationship holds:

$$P\left(\frac{\sigma_\nu}{\sqrt{1-\rho^2}}\,\nu \leq \log(\bar{\epsilon}_i) + \frac{d_i}{2}\right) = \frac{i}{N_\epsilon} \text{ for } i = 1, \cdots, N_\epsilon - 1, \tag{56}$$
$$\text{where } d_i = \log(\bar{\epsilon}_{i+1}) - \log(\bar{\epsilon}_i) \text{ and } \nu \sim N(0,1).$$

Note that the variance of $\log(\epsilon_t)$ in the stationary distribution is given by $\left(\frac{\sigma_\nu}{\sqrt{1-\rho^2}}\right)^2$. Thus, the cumulative probability at each grid point is approximately equal to $\frac{i}{N_\epsilon}$ for $i = 1, \cdots, N_\epsilon - 1$. The number of grid points, N_ϵ, is chosen to be an odd number, and the grid is symmetric around the middle point, $\log\left(\epsilon_{\frac{1}{2}(N_\epsilon+1)}\right) = 0$. Once the grid points $\{\bar{\epsilon}_i\}_{i=1,\cdots,N}$ are chosen, the transition probability is computed as:

$$P(\epsilon_t = \bar{\epsilon}_i | \epsilon_{t-1} = \bar{\epsilon}_j) = P\left(\log(\bar{\epsilon}_i) - \frac{d_{i-1}}{2} \leq \rho \log(\bar{\epsilon}_j) + \sigma_\nu \nu \leq \log(\bar{\epsilon}_\epsilon) + \frac{d_i}{2}\right) \text{ for } i = 2, \cdots, N_\epsilon - 1,$$

while the transition probability at the first and last grid point is computed as:

$$P(\epsilon_t = \bar{\epsilon}_1 | \epsilon_{t-1} = \bar{\epsilon}_j) = P\left(\rho\bar{\epsilon}_j + \sigma_\nu \nu \leq \bar{\epsilon}_1 + \frac{d_1}{2}\right), \tag{57}$$

$$P(\epsilon_t = \bar{\epsilon}_{N_\epsilon} | \epsilon_{t-1} = \bar{\epsilon}_j) = P\left(\bar{\epsilon}_{N_\epsilon} - \frac{d_{N_\epsilon - 1}}{2} \leq \rho\bar{\epsilon}_j + \sigma_\nu \nu\right). \tag{58}$$

Using this method, the probability at ϵ_3 in case of $N_\epsilon = 5$ is 0.22, while in the standard method, the probability at ϵ_3 is 0.37 if a multiple of two of the unconditional standard deviation is chosen for ϵ_{ϵ_N}.

B.3.3 Note on Stationary Distribution

A stationary distribution is computed by simulation. The stationary distribution Ψ^* is defined over state variables $h := (n, \epsilon, o, \hat{\theta})$.

Given policy functions obtained from the value function iteration, it is straightforward to simulate an evolution of state variables over the firm's life cycle. This is done by drawing the initial state $h_0 = (n_0, \epsilon_0, o_0, \hat{\theta})$ and realization of shocks (ϵ_t and innovation outcome) over the life cycle. The control variables $(\ell_t, k_t, I_t, b_t, d_t)$ are determined by the policy functions conditional on the firm's state variables.

Although firm age is not a part of state variables, the stationary distribution over the state variable h can be expressed as a sum of distributions at each age:

$$\Psi^*(h) = \sum_{a=0}^{\bar{a}} \Psi^*(h; a), \tag{59}$$

where a denotes firm age from 0 to $\bar{a}$, and $\bar{a}$ is the maximum firm age. A large value for $\bar{a}$ is chosen (e.g., $\bar{a} = 100$). Note that a stationary distribution is typically computed by drawing an arbitrary initial distribution Ψ^0 and sequentially updating the distribution until it converges based on a transition function P:

$$\Psi^{j+1} = P(\Psi^j). \tag{60}$$

Here, using the distribution conditional on age, the distribution at iteration j is expressed as:

$$\begin{aligned} \Psi^j &:= \sum_{a=0}^{\bar{a}} \Psi^j(h; a) \\ &= M^e \bar{\psi} + \sum_{a=1}^{\bar{a}} P_{a-1 \to a}\left(\Psi^{j-1}(h; a-1)\right), \end{aligned} \tag{61}$$

where $\bar{\psi}$ denotes the distibution of entrants. A transition probability $P_{a-1\to a}$ computes the distribution $\Psi^j(h;a)$ over h at age a at iteration j from the distribution $\Psi^{j-1}(h;a-1)$ over h at age $a-1$ at iteration $j-1$. The distribution Ψ^j at age a can be computed by a simulation of many firms (e.g., 10,000 firms) based on the distribution at age $a-1$. If firms exit the market, all of their control variables are set to zero from the period onwards. The measure of the entrants is given by M^e and is a constant in the steady state. By taking a large j that satisfies $j > \bar{a}$, the stationary distribution Ψ^* over h is expressed as:

$$\Psi^* = M^e\bar{\psi} + P_{0\to 1}\left(M^e\bar{\psi}\right) + P_{0\to 2}\left(M^e\bar{\psi}\right) + \cdots + P_{0\to \bar{a}}\left(M^e\bar{\psi}\right). \tag{62}$$

This implies that the stationary distribution can be derived as a sum of distributions across all ages from $a=0$ to $a=\bar{a}$, given the initial distribution of entrants ψ. The distribution of each age group $P_{0\to a}\left(M^e\bar{\psi}\right)$ can be simulated given the initial distribution of entrants.